< 1-10-1-1-1_ANT_nCA_GLOBAL >

Rev_0.3

B17	EAM63971201
B20	EAM63470401
B28	EAM63491001

LG Electronics

Schematic1

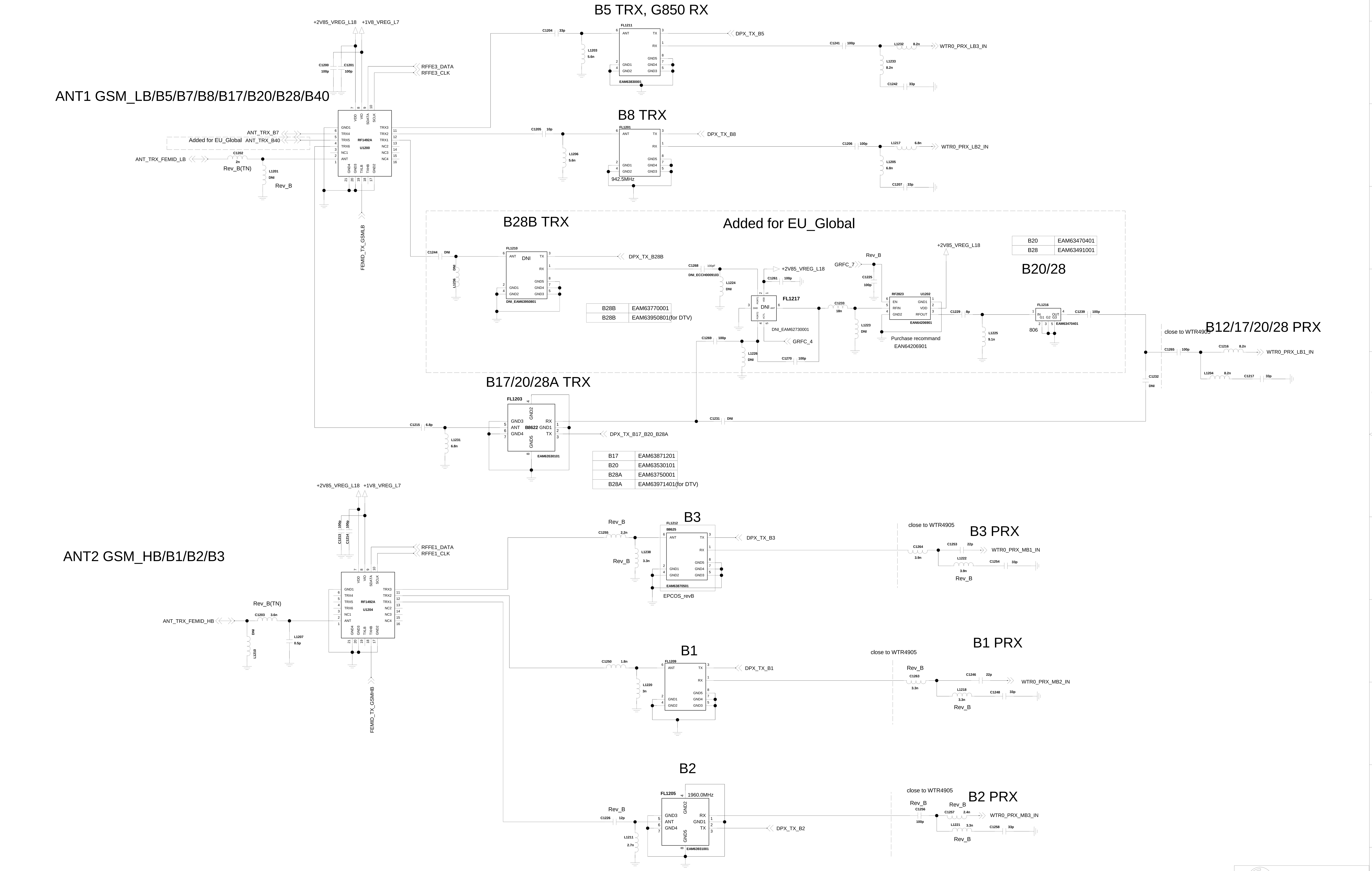
A2 PDM NUMBER Rev

User Name

Drawing Date

LGE Internal Use Only

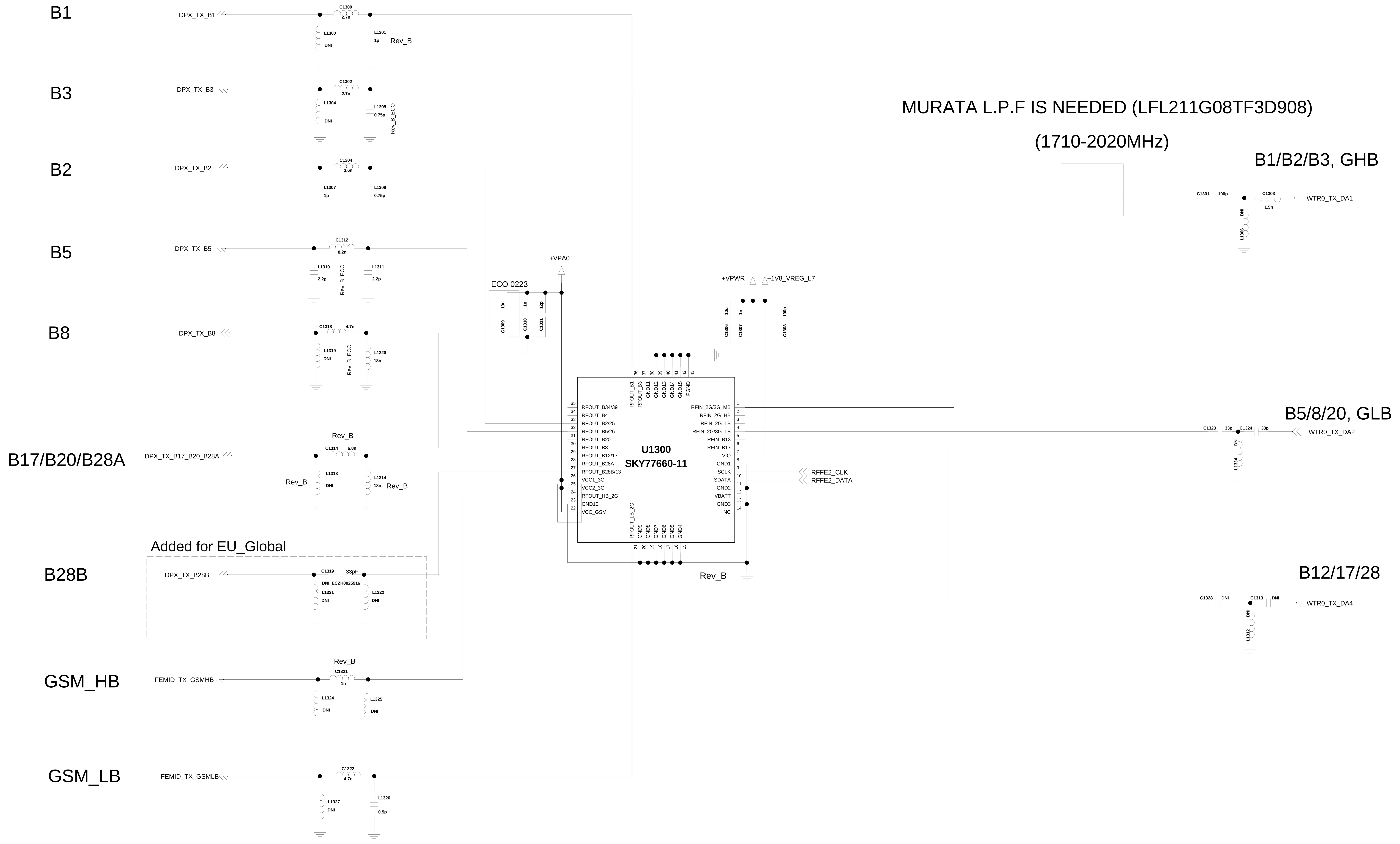
< 1-10-1-2-1_SP8T_SP8T_nCA_CA> Rev_0.3



LGE Internal Use Only

LG Electronics
Schematic1
A2 PDM NUMBER Rev
User Name Drawing Date

< 1-11-1-3-1_uMMPA_APT_ACPM-7788 > Rev_0.1

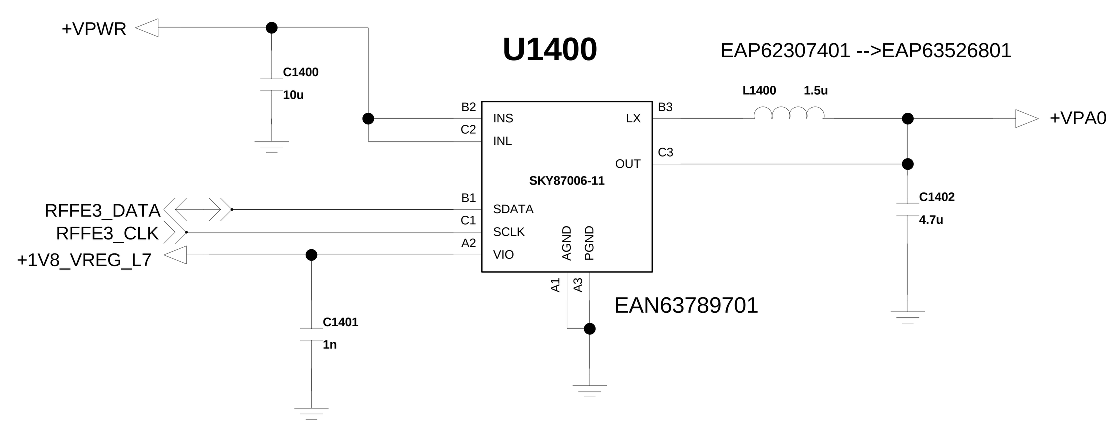


LGE Internal Use Only

16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

< 1-2-4-3_PT_APT_QFE2101 >

Rev_0.3 -> SKY87006



LGE Internal Use Only

LG Electronics

TITLE Schematic1

SIZE A2 | PDM NUMBER | Rev

DRAWN BY User Name | SHEET 2 of 2 | Drawing Date

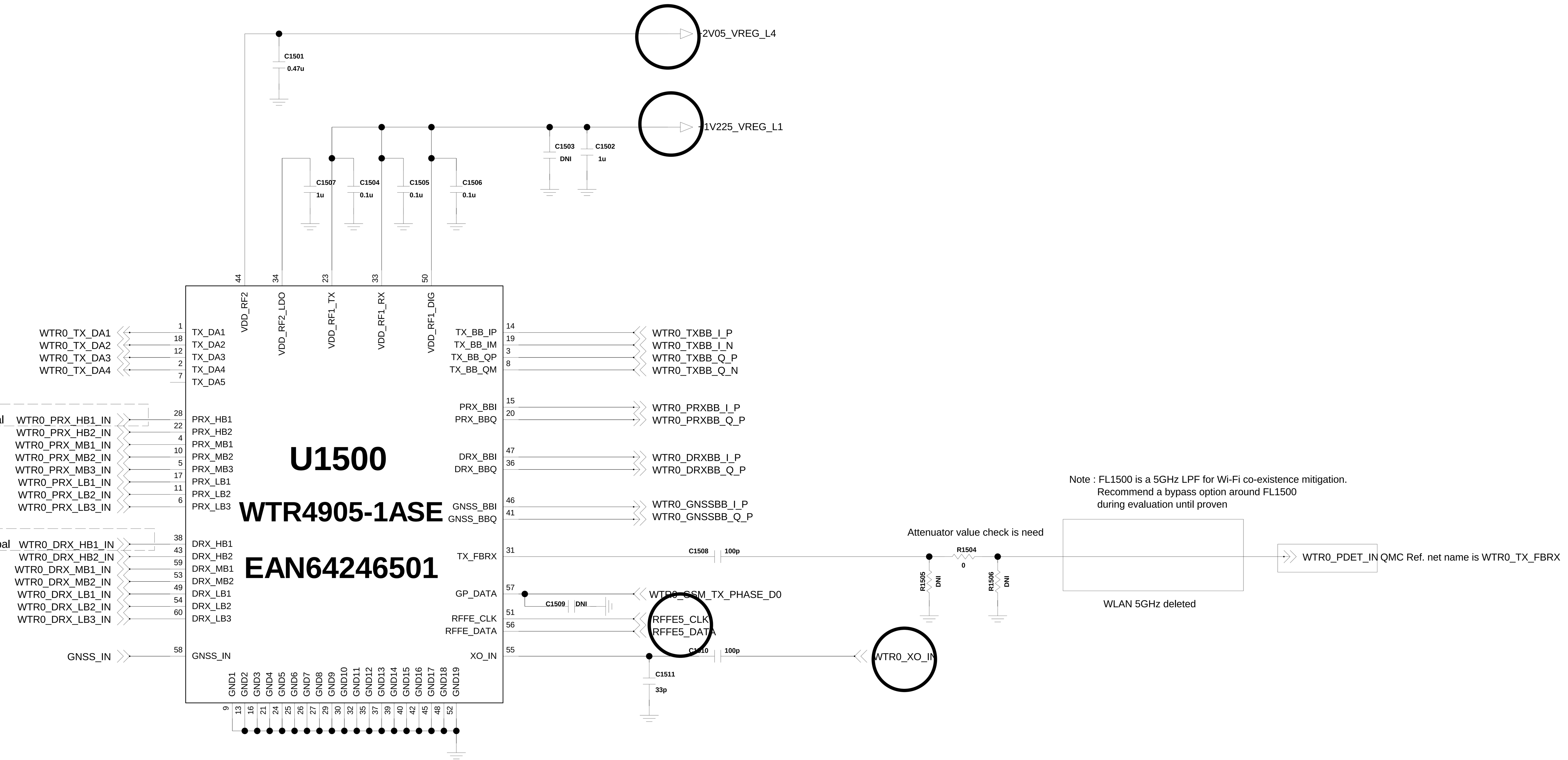
16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

<1-8-5-1_WTR4905>

Rev_0.5

Tx Table

TX_DA1	B1/2/3, GSM_HB
TX_DA2	B5/8/20, GSM_LB
TX_DA3	B7, B40
TX_DA4	B17/28
TX_DA5	-



ECO_0213

LG Electronics

TITLE Schematic1

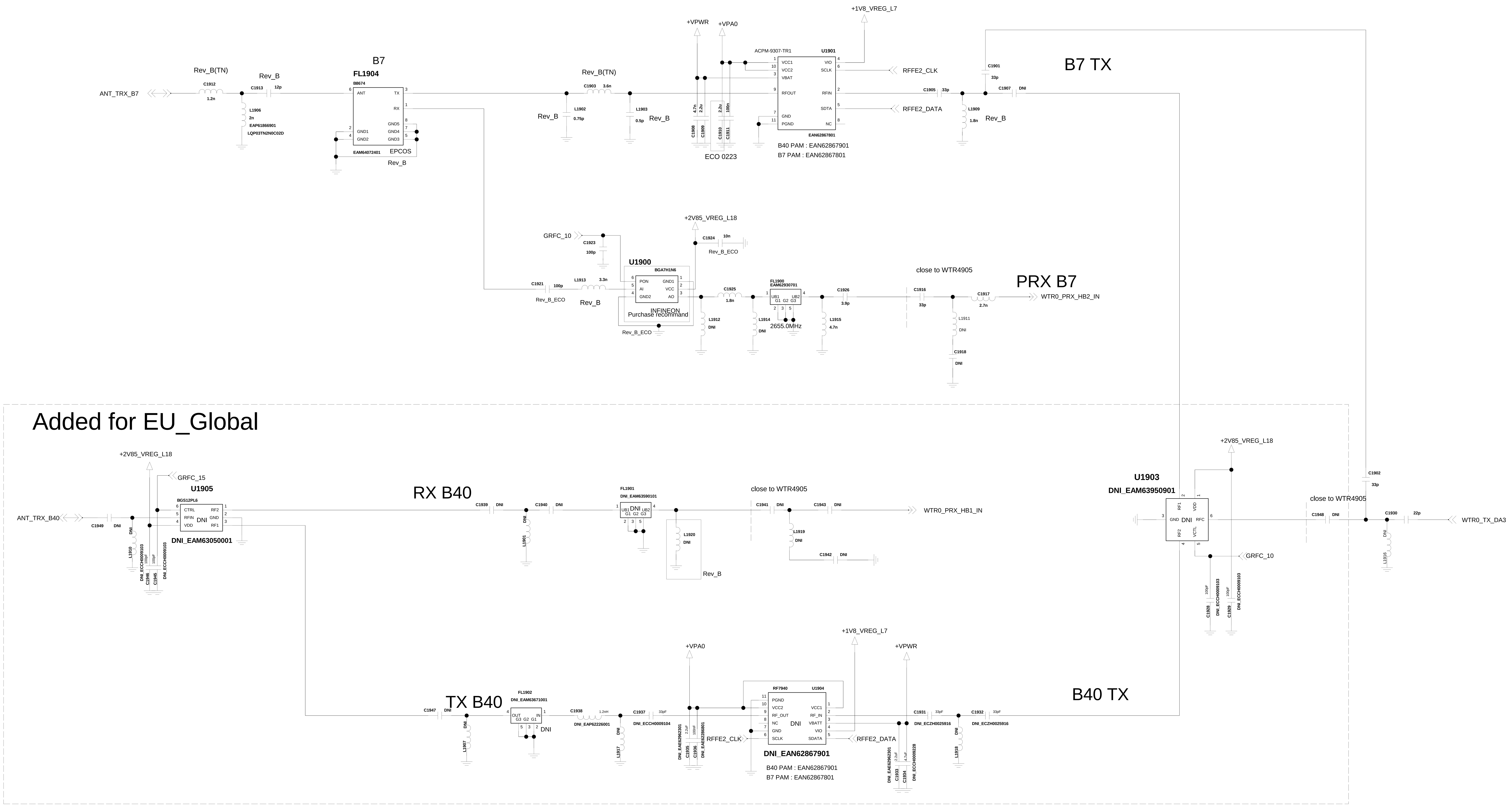
A2 PDM NUMBER Rev

User Name 3 of 3 Drawing Date

LGE Internal Use Only

< 1-3-6-xx_TRX_B40_B7 >

Rev_0.4



LGE Internal Use Only

LG Electronics

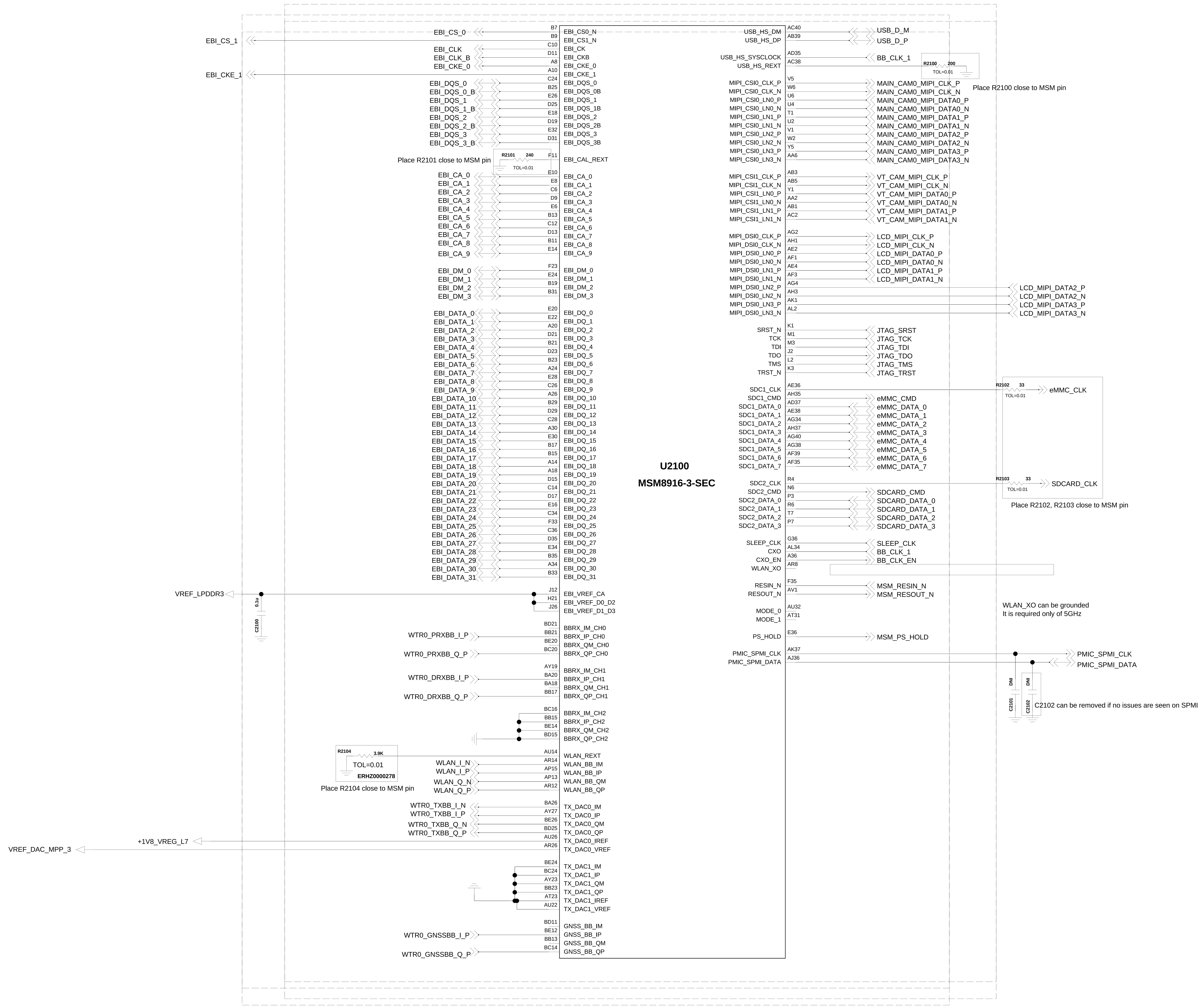
Schematic1

A2 | PDM NUMBER | Rev

User Name | Drawing Date

< 2-1-5-3-1_MSM8916-3-SEC_CONTROL > REV_0.3

CCDS CARD Information	
Release Date	June. 27 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	

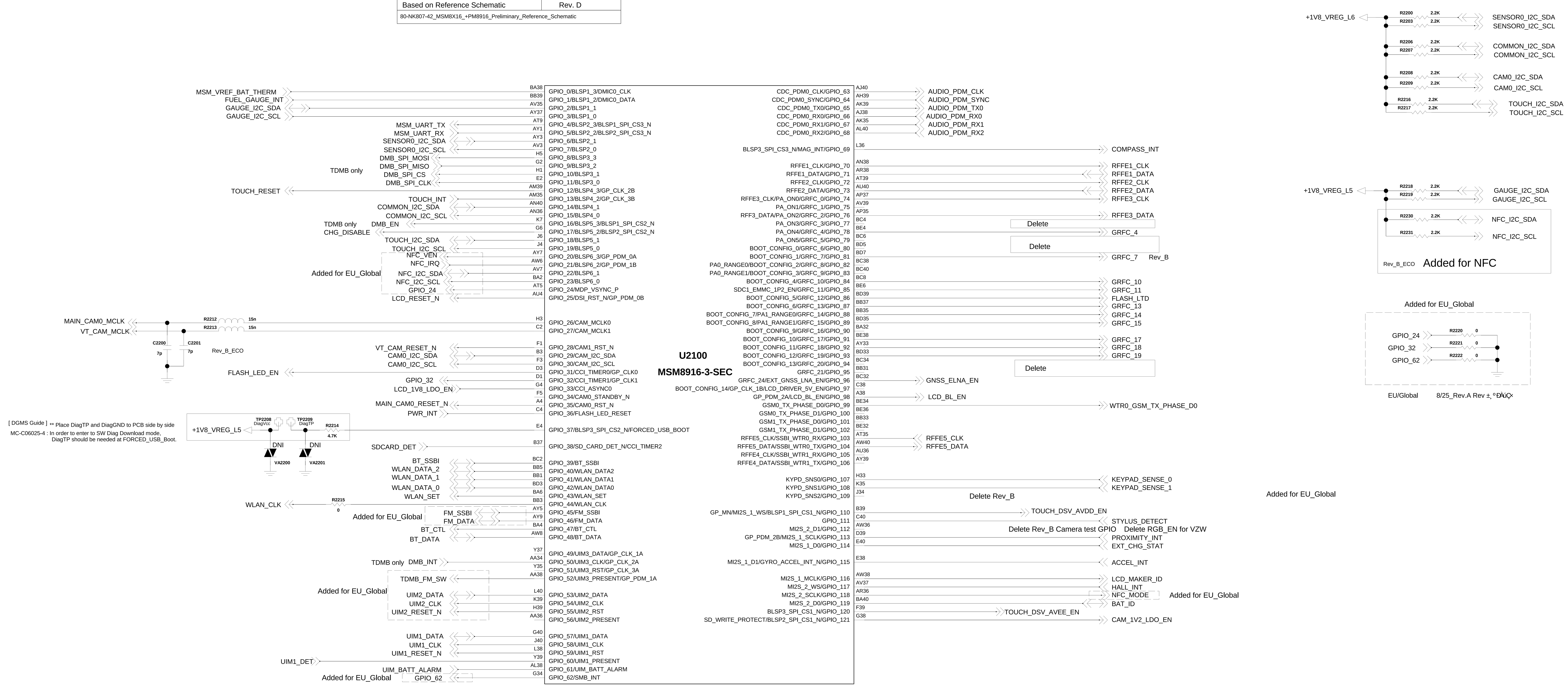


LGE Internal Use Only

LG Electronics		
TITLE	Schematic1	
SIZE A2	PDM NUMBER	Rev
SHEET 2 of 2	Drawing Date	
DRAWN BY	User Name	

< 2-1-5-3-2_MSM8916-3-SEC_GPIO >   Rev_0.3

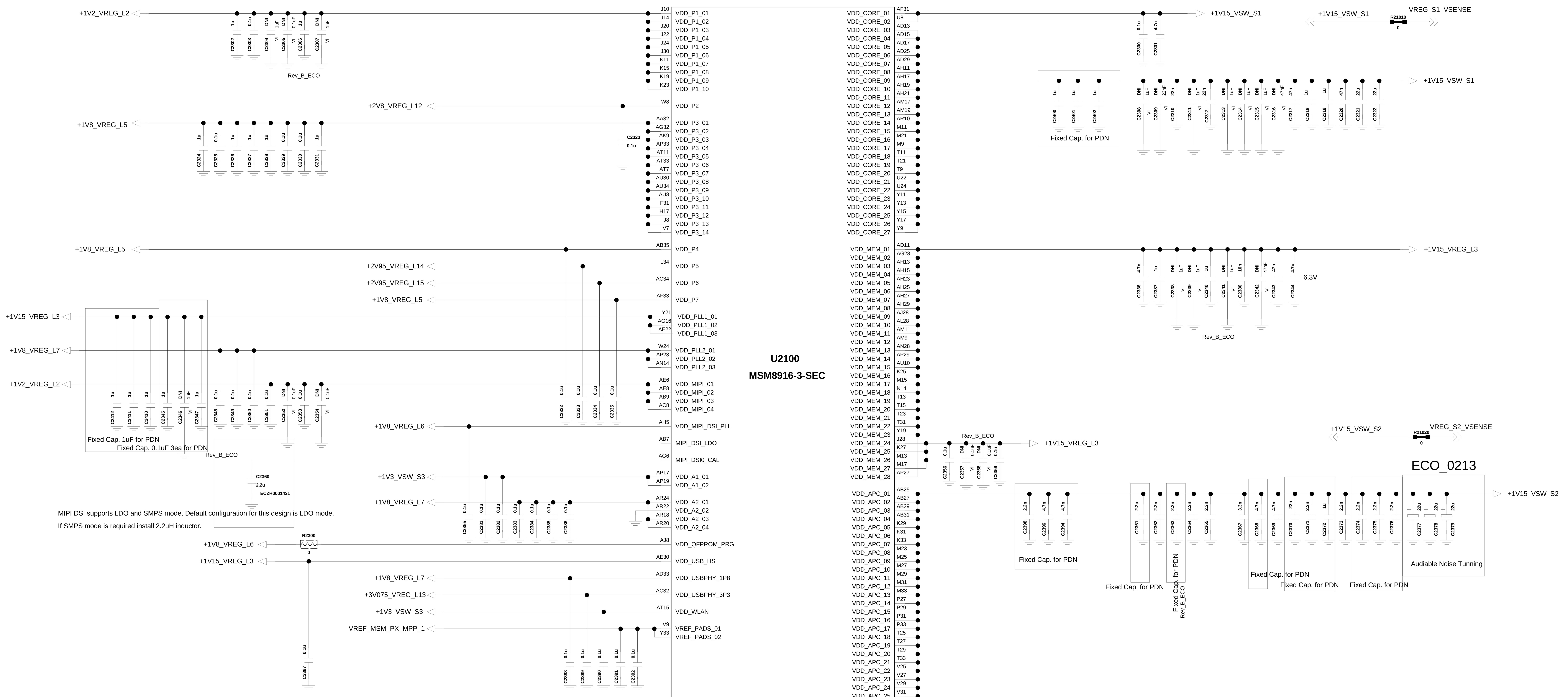
CCDS CARD Information	
Release Date	Sep. 24 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	



LGE Internal Use Only

< 2-1-5-3-3_MSM8916-3-SEC_POWER > Rev_0.3

CCDS CARD Information	
Release Date	Sep. 24 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	

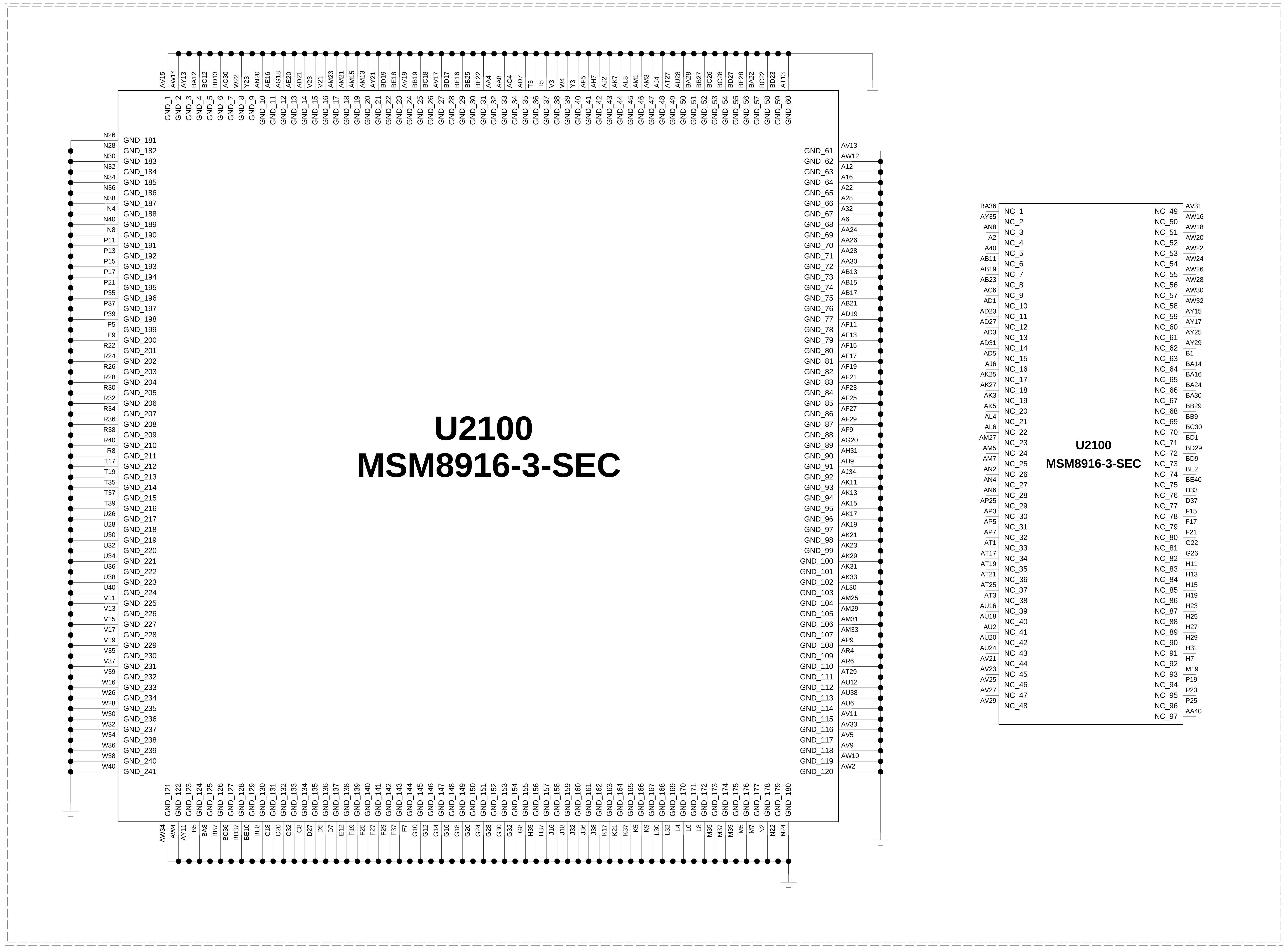


LGE Internal Use Only

LG Electronics		
TITLE	Schematic1	
SIZE A2	PDM NUMBER	REV Rev
SHEET 4 of 4	Drawing Date	DRAWN BY User Name

< 2_1_5_3_4_MSM8916_3_SEC_GND_NC > REV_0.3

CCDS CARD Information	
Release Date	Sep. 24 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	



LGE Internal Use Only

LG Electronics

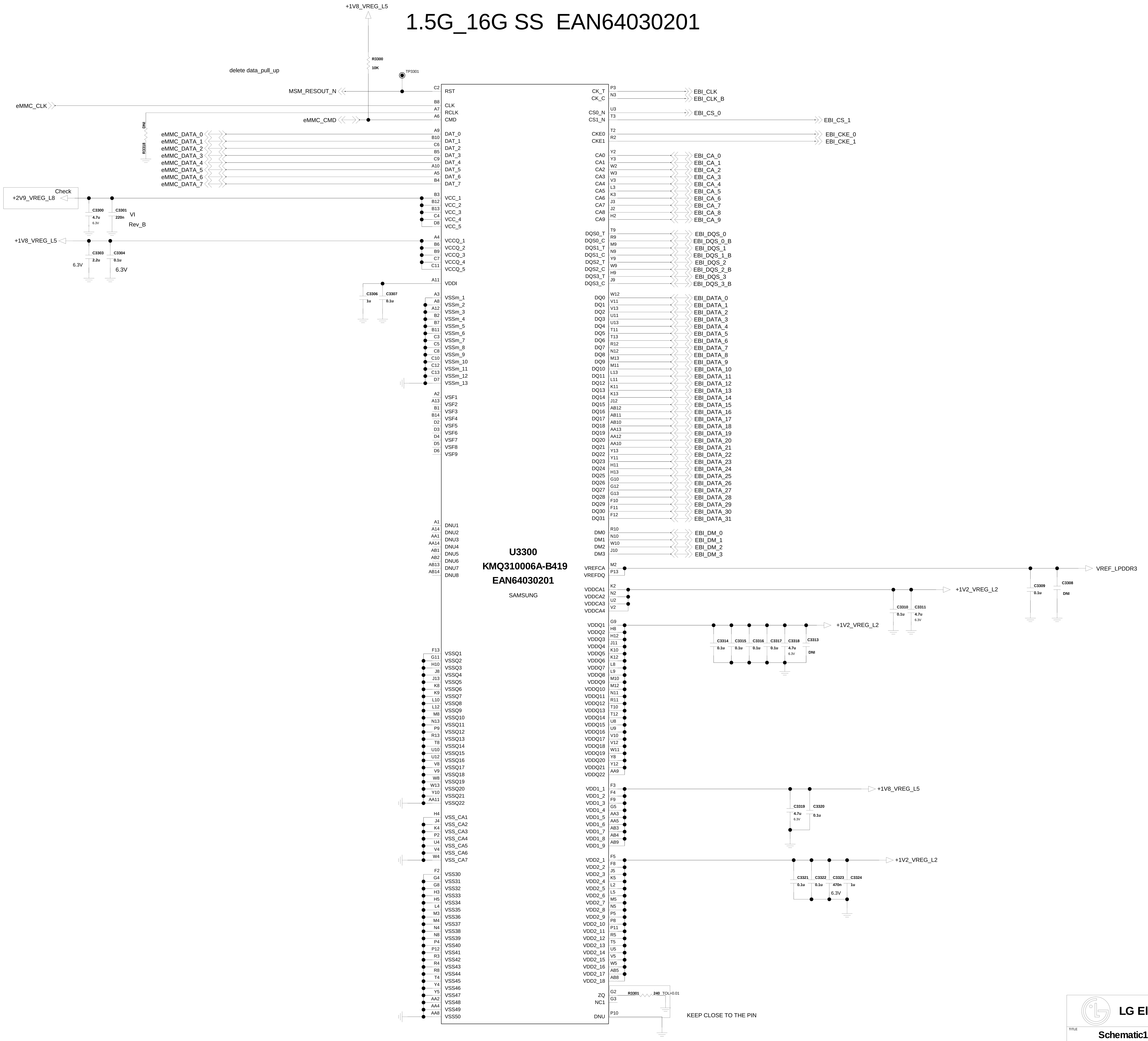
Schematic1

A2 PDM NUMBER Rev

User Name Drawing Date

< 3-3-3-3-1_MCP_eMMC_5_0_8G_DDR3_12G_Samsung > Rev_0.3

1.5G_16G SS EAN64030201



LGE Internal Use Only

LG Electronics

Schematic1

SIZE	DWG NO	REV
A2	PDM NUMBER	Rev

SHEET 1 of 29

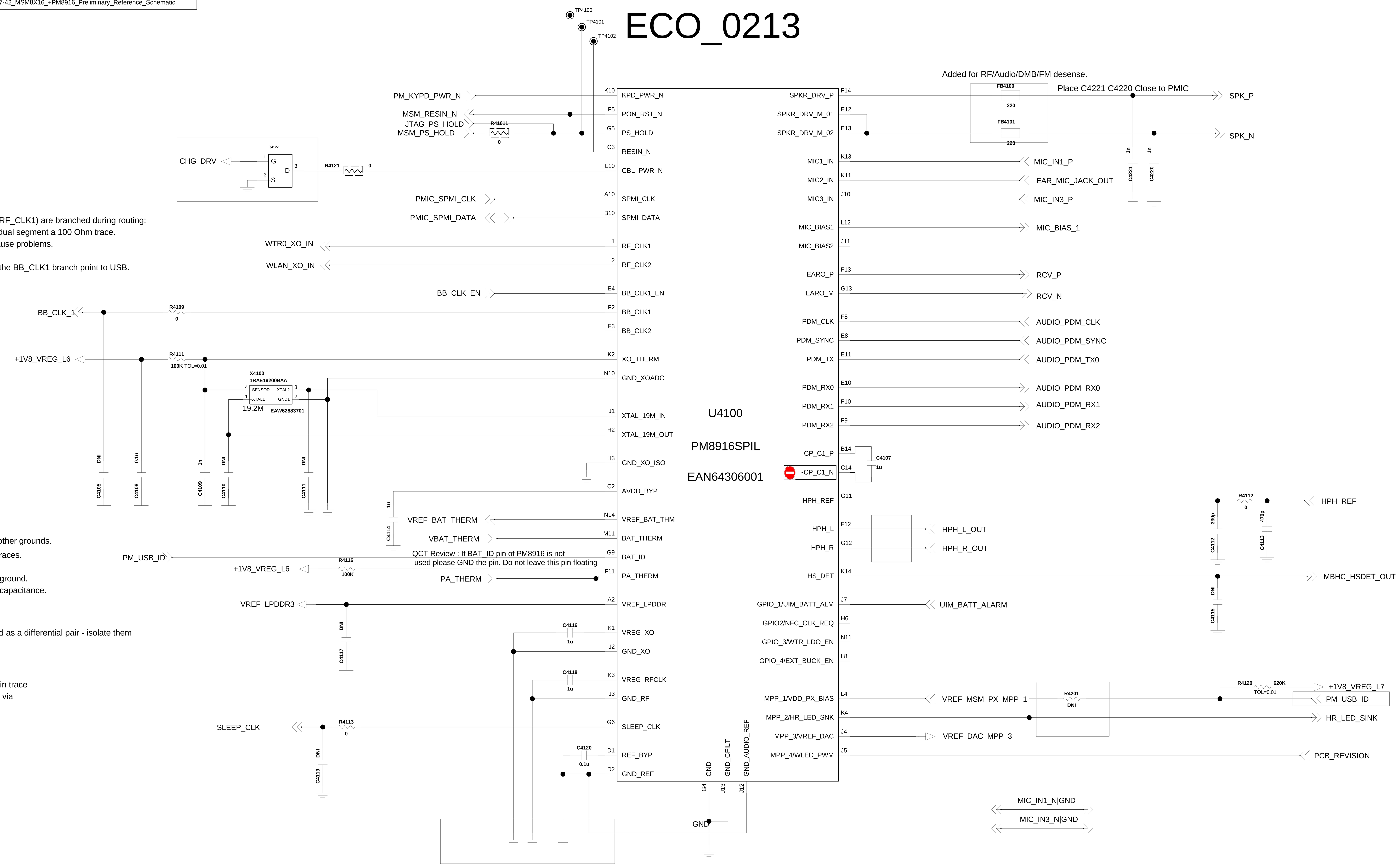
DRAWN BY: User Name

Date: Drawing Date

<4-1-9-1_PM8916_DATA> Rev_1.0

CCDS CARD Information	
Release Date	June. 27 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	

ECO_0213



When the PMIC¡¯s 19.2 MHz clock signals (BB_CLK1 and RF_CLK1) are branched during routing:
¡¤Make the common route a 50 Ohm trace, and each individual segment a 100 Ohm trace.
¡¤Match the branch lengths; even a 5 mm difference can cause problems.
¡¤Keep branches as short as possible.
¡¤If these routing guidelines cannot be met, add a buffer at the BB_CLK1 branch point to USB.

A ¡®moat¡¯ cleared of metal that isolates XO ground from other grounds.
XO ground points connect to GND_XOADC pin using thin traces.
GND below in/out lines for improved isolation
Via to main ground plane under the XO_THERM capacitor ground.
Maintain clearance below in/out pads for reduced parasitic capacitance.
Locate the crystal 1 cm from the PMIC
Use thin traces for high thermal resistance
XTAL_19M_IN and XTAL_19M_OUT should NOT be routed as a differential pair - isolate them
Isolate from other signals (no parallel routing)
R and C close to crystal
Ground connections
Connect crystal GND to PMIC GND_XOADC pin using a thin trace
Connect GND_XOADC pin to main GND using a dedicated via

	VZW	Other
MPP_2	VZW_CHG	HR_LED_SINK
R4201	0	DNI
R4202	DNI	0

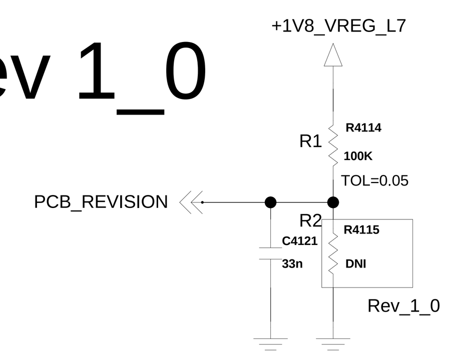
GND_REF must connect directly to C4120 and then to main GND
without any other intermediate connections to GND fills

GND_XO and GND_RF must connect directly to C4118 and C4116 respectively
and then connect to main GND without any other intermediate connections to GND fills

It is recommended that the handset microphone ground or negative
are routed differentially with microphone input trace
and ground to the same ground as GND_CFILT
80-NK808-121 PM8916 CODEC BANDGAP REFERENCE AND MICROPHONE PCB CONNECTIONS TECHNICAL MEMO

PCB REVISION

Rev 1_0



Rev.	R1	R2	Voltage
A	100K	20K	0.300
B	100K	27K	0.383
C	100K	36K	0.48
D	100K	51K	0.680
E	100K	75K	0.771
F	100K	100K	0.900
G	100K	130K	1.017
1.0	100K	DNI	1.800
1.1	DNI	100K	0.000
MKT	100K	200K	1.2

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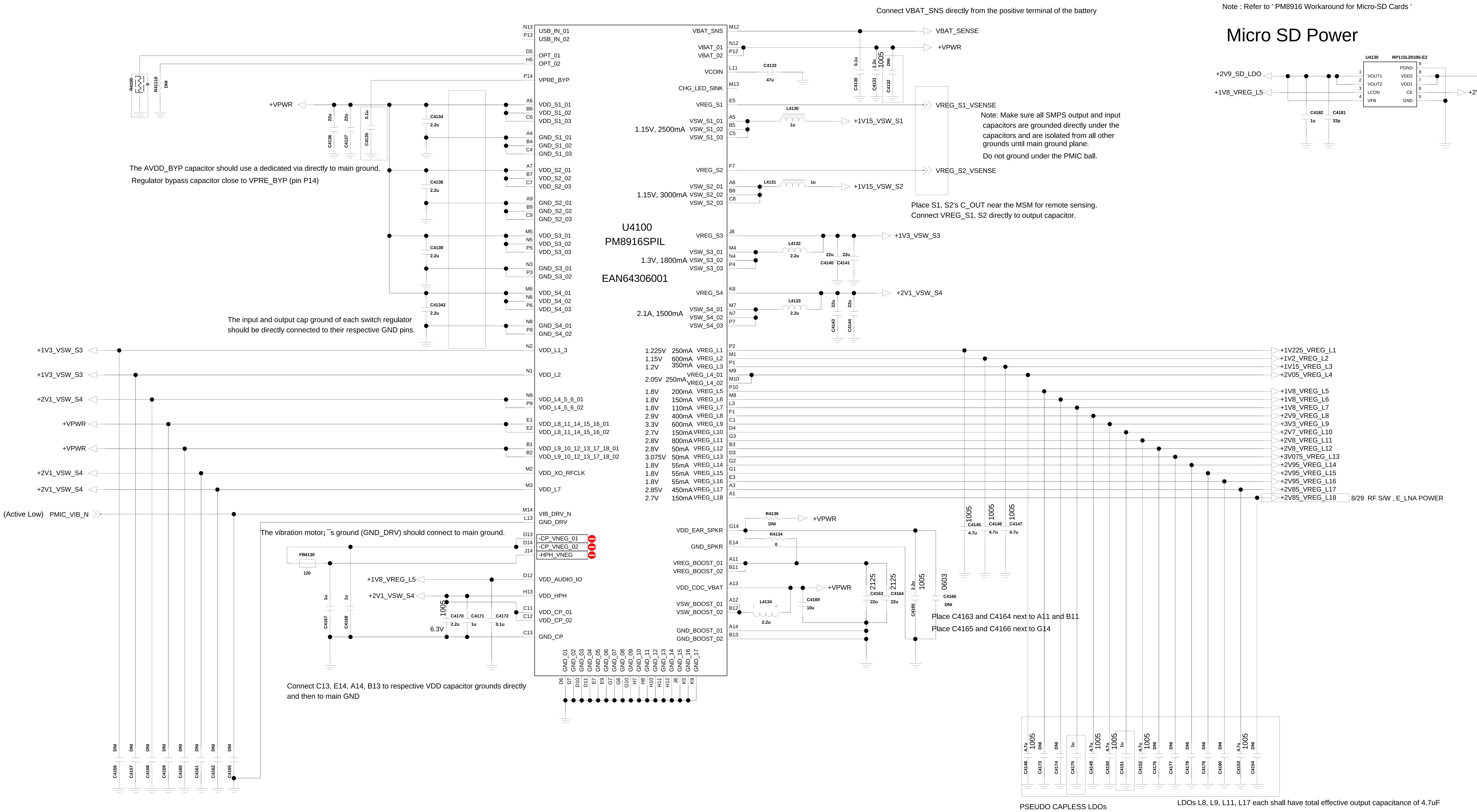
LG Electronics	
TITLE	Schematic1
SIZE A2	DWG NO PDM NUMBER
REV	Rev
DRAWN BY	User Name
SHEET 28 of 27	Date Drawing Date

<4-1-9-2_PM8916_POWER> REV_1.0

CCDS CARD Information	
Release Date	June. 27 2014
Based on Reference Schematic	Rev. D
80-NK807-42_MSM8X16_+PM8916_Preliminary_Reference_Schematic	

ECO_0213



LDOs L8, L9, L11, L17 each shall have total effective output capacitance of 4.7uF

Rest of the LDOs shall each have total effective output capacitance of 1.0uF

The Pseudo capless LDOs must comply to following tolerable routing requirement:

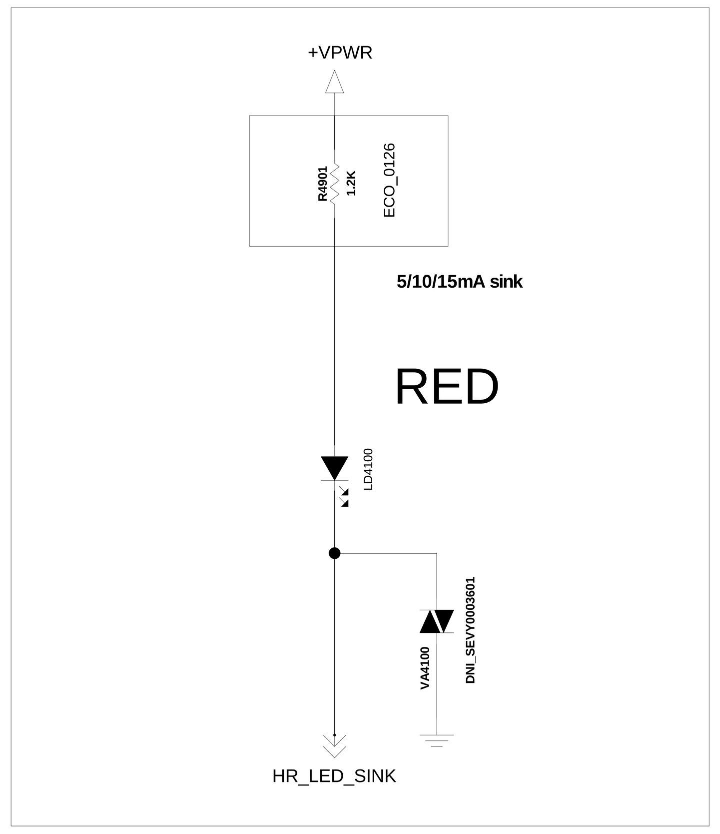
P50: R<400mOhms, L<45nH
P150: R<200mOhms, L<20nH
P300: R<150mOhms, L<18nH
P600: R<150mOhms, L<18nH

Function	Circuit Type	Default Voltage	Specified Range	Programmable Range	Rated Current	Default On
S1	HF_SMPS	1.15V				Y
S2	HF_SMPS	1.15V				N
S3	HF_SMPS	1.3V				Y
S4	HF_SMPS	2.1V				Y
L1	NMOS LDO	1.225V				N
L2	NMOS LDO	1.15V				Y
L3	NMOS LDO	1.2V				Y
L4	PMOS LDO	2.05V				N
L5	PMOS LDO	1.8V				Y
L6	PMOS LDO	1.8V				N
L7	PMOS LDO	1.8V				N
L8	PMOS LDO	2.9V				N
L9	PMOS LDO	3.3V				N
L10	PMOS LDO	2.7V				N
L11	PMOS LDO	2.8V				N
L12	PMOS LDO	2.8V				N
L13	PMOS LDO	3.075V				N
L14	PMOS LDO	1.8V				N
L15	PMOS LDO	1.8V				N
L16	PMOS LDO	1.8V				N
L17	PMOS LDO	2.85V				N
L18	PMOS LDO	2.7V				N
VREG_XO	PMOS LDO	2.9V				
VREG_RF_CLK	PMOS LDO	2.95V				

LGE Internal Use Only

<4-6-1_LED_Driver_IC_LP5521>

REV.1.0



For VZW

<4-8-2_Fuel_Gauge_MAX17050>

Rev_1.0

Delete Fuel Gauge MAX17050 for VZW

For VZW

<4-3-4_OVP_R5560Z011A-TL-F>

15/02/13 Change for low cost

OVP Default 6.8V(C1 Pin Connet to GND)

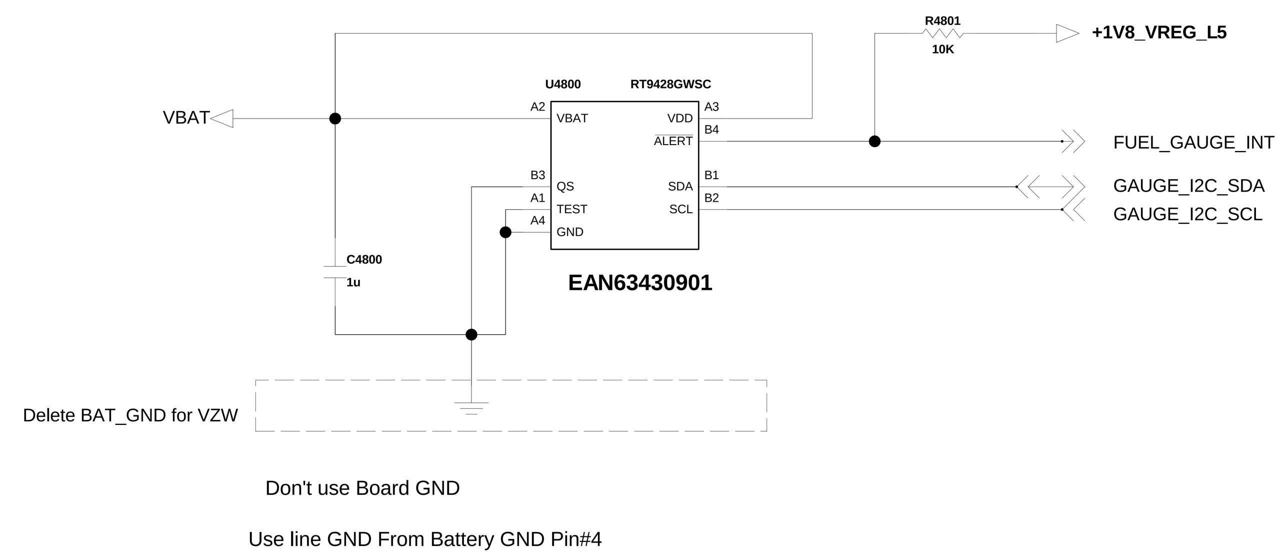
REV_1.0

Delete OVP for VZW

<4-8-1_Fuel_Gauge_RT9428>

Only use line VBAT1 from Battery Connector.

CLOSE TO Battery CONNECTOR

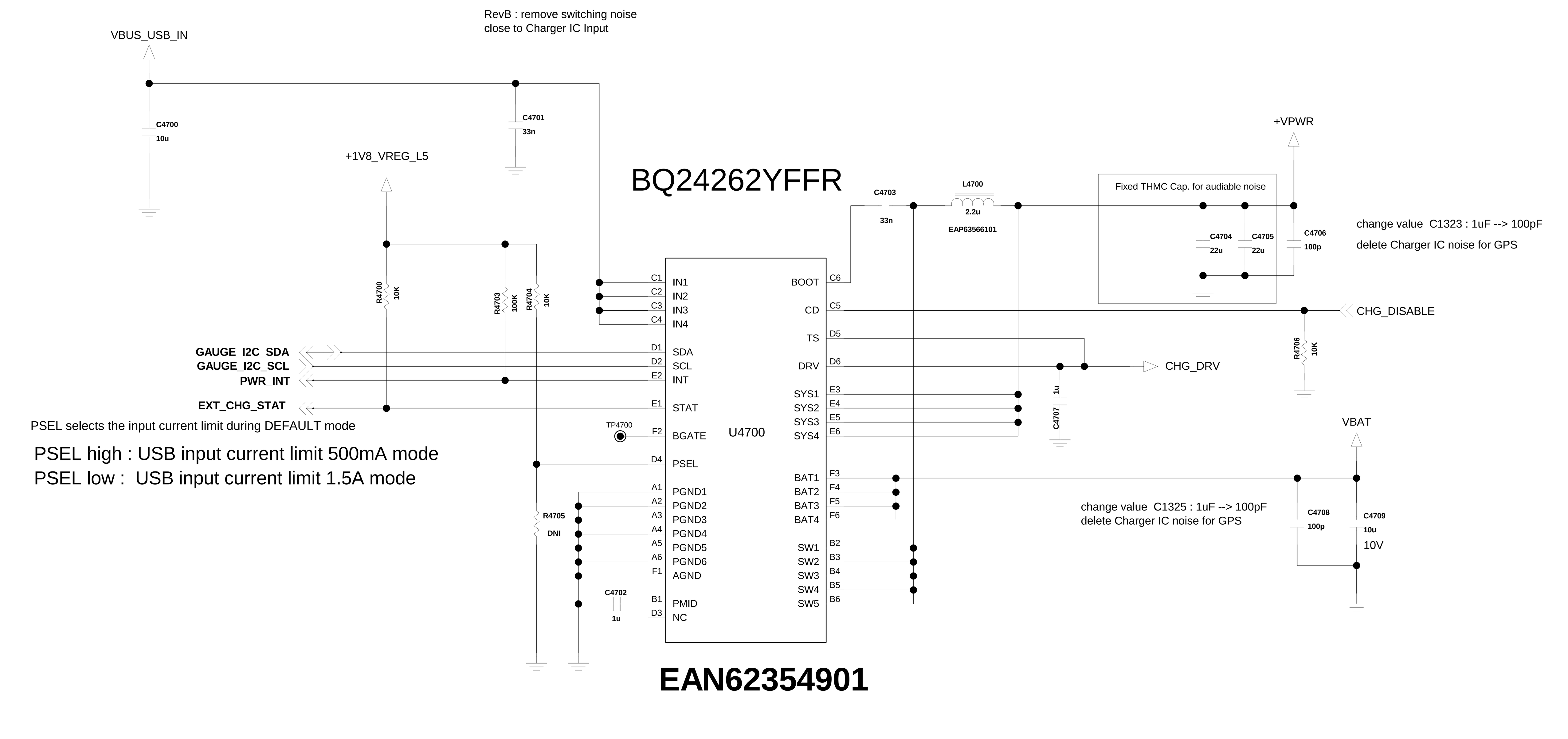


For VZW

Delete BQ24292 For VZW

CHARGING IC

Add Charger IC _130808



LG Electronics

Schematic1

A2 PDM NUMBER Rev

33 of 30

User Name Drawing Date

LGE Internal Use Only

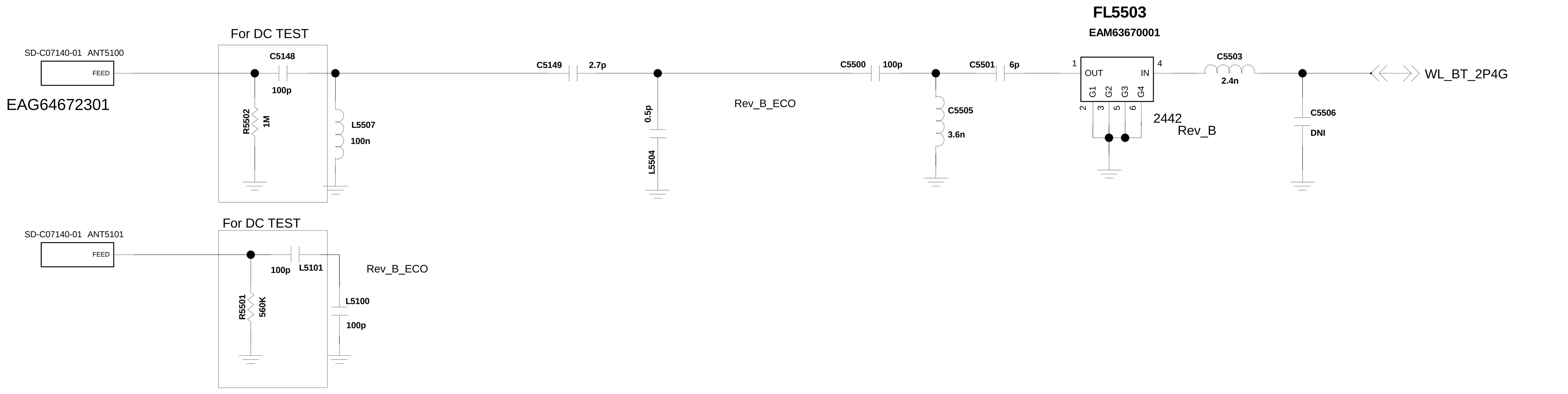
< 5-1-1-17_BT_Wi-Fi_WCN3660B > Rev_1.0

ANT (BT/WIFI)

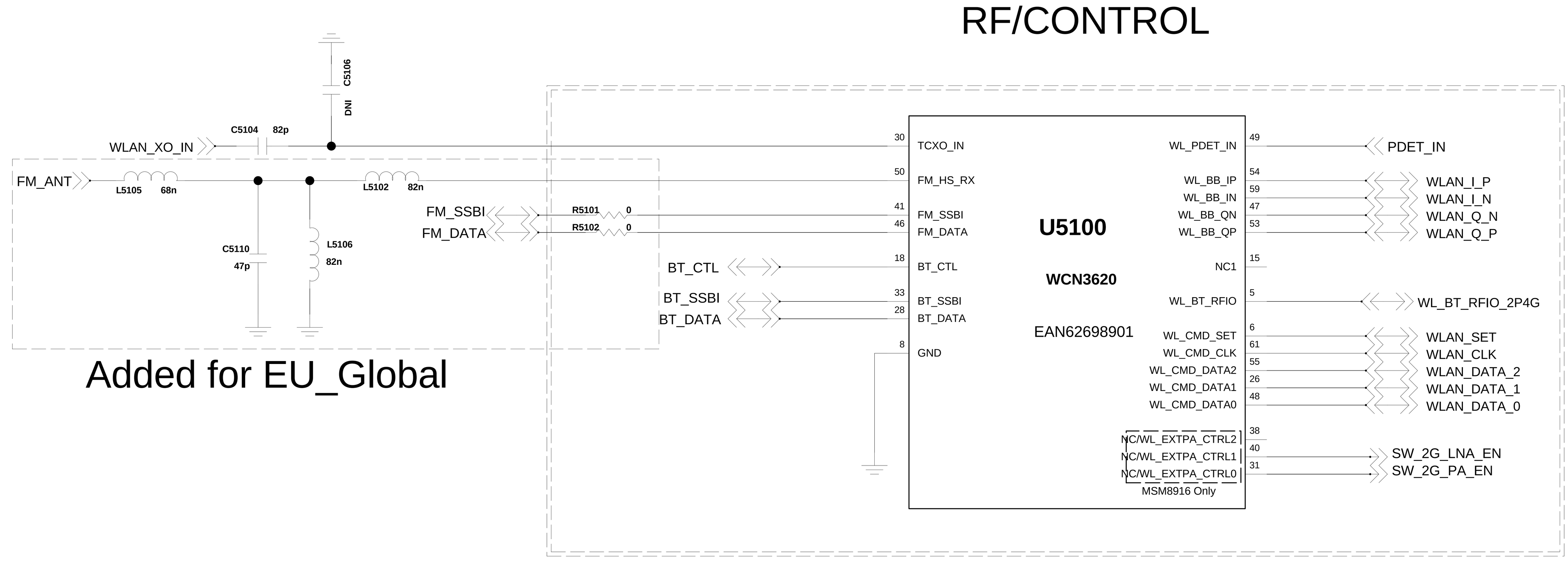
BT/WiFi RF Filter

RF Filter

Library update



RF/CONTROL

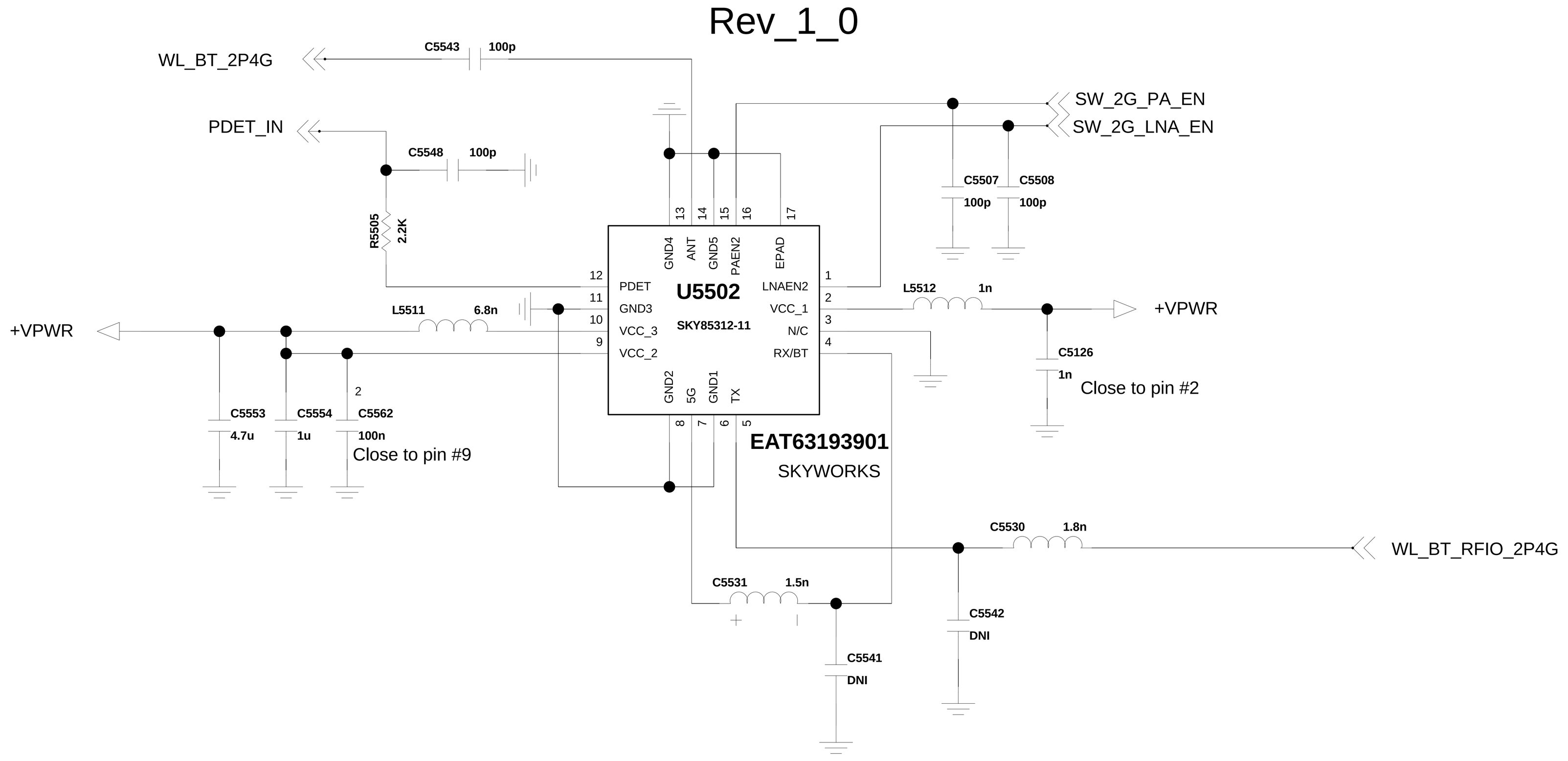


Option : 2G FEM (pin compatible)

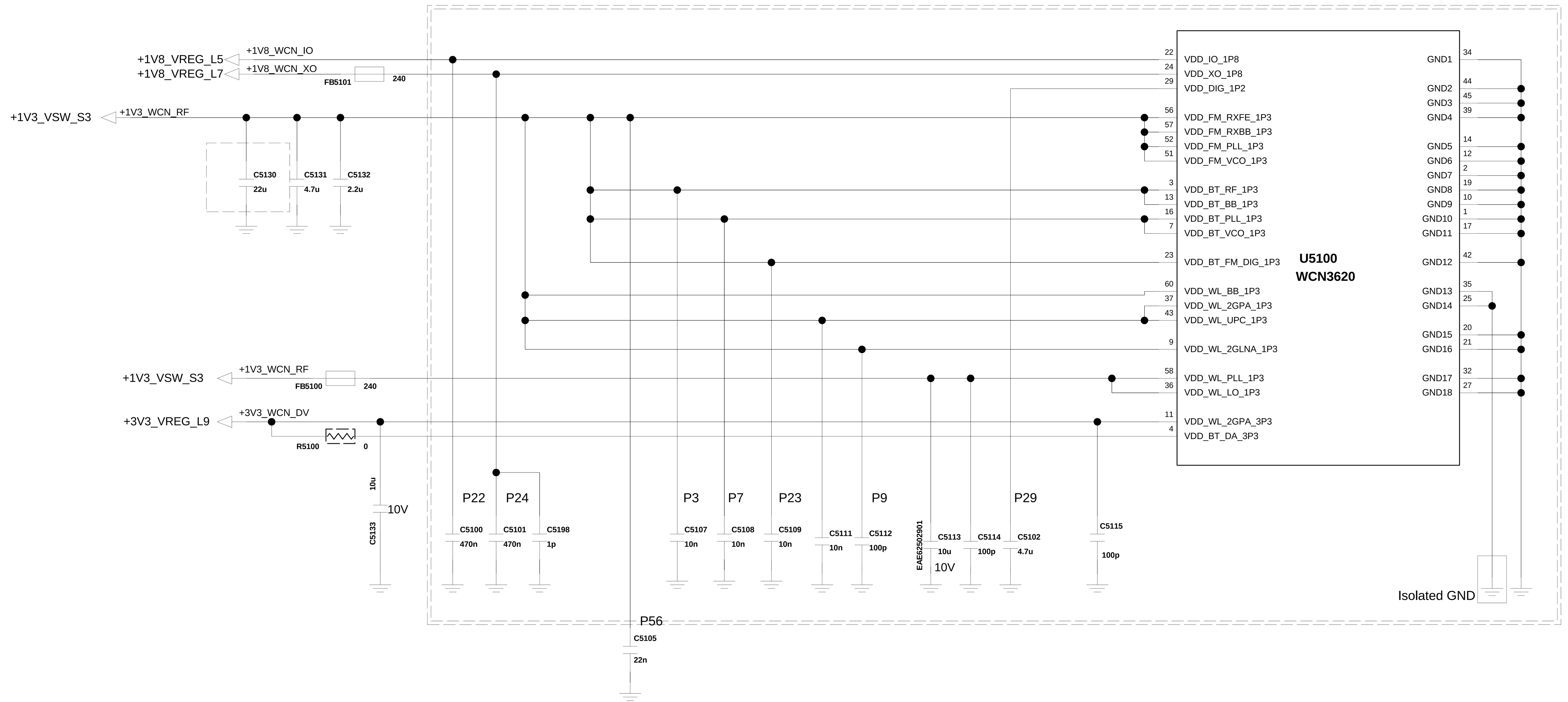
1. SKY85312-11 2.RFFM8216

2G FEM(SP3T +2G PA + LNA) + switch

Rev_1_0



POWER/GND



LG Electronics

TITLE

A2

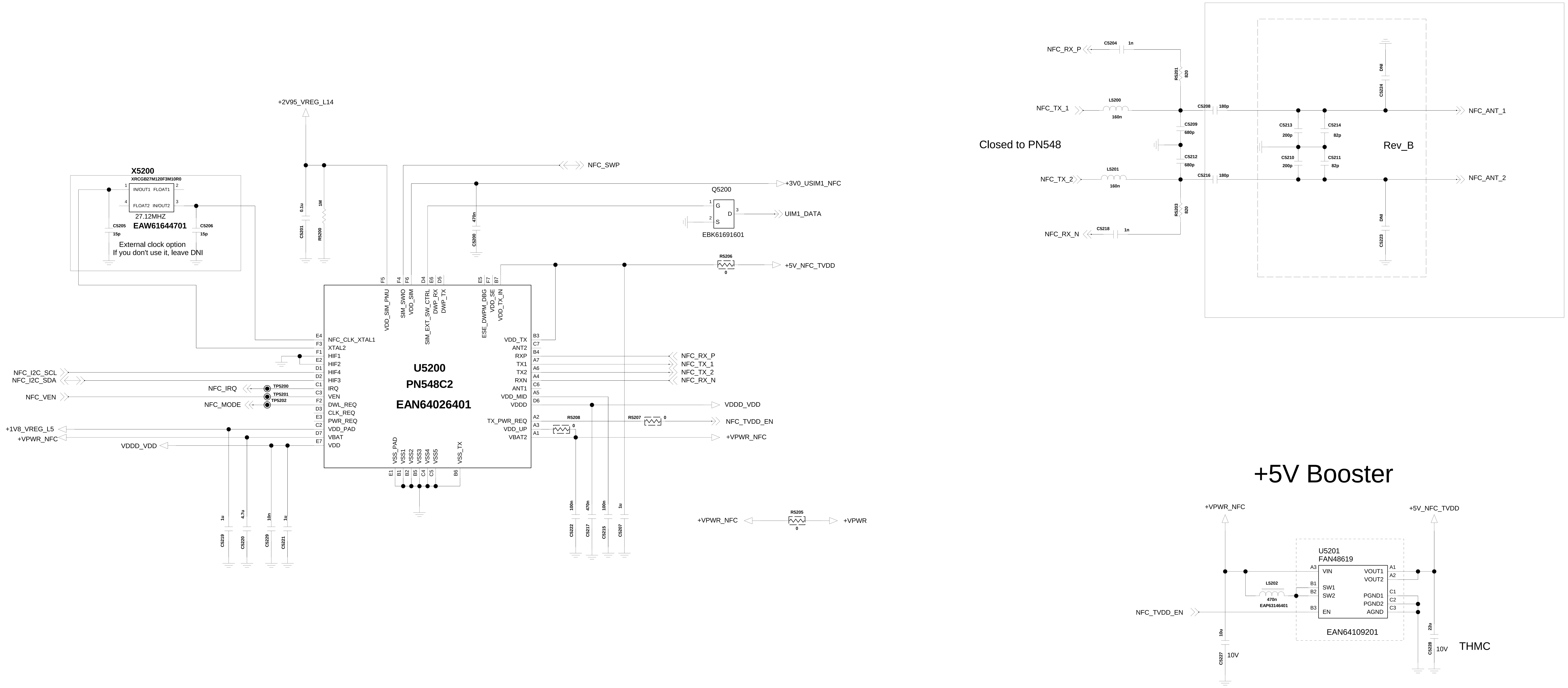
heesoo.kim

LGE Internal Use Only

Added for NFC Rev_B_ECO

<NFC_PN548>

NFC Antenna



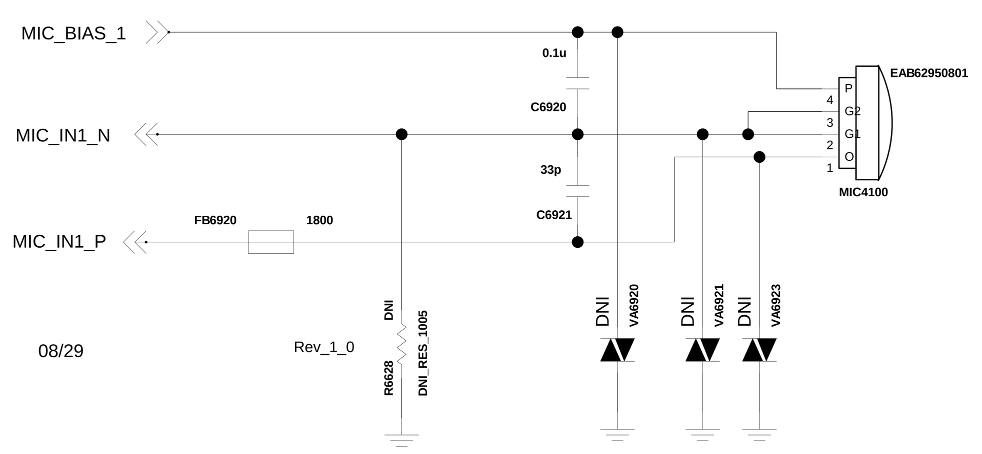
LG Electronics

Schematic1

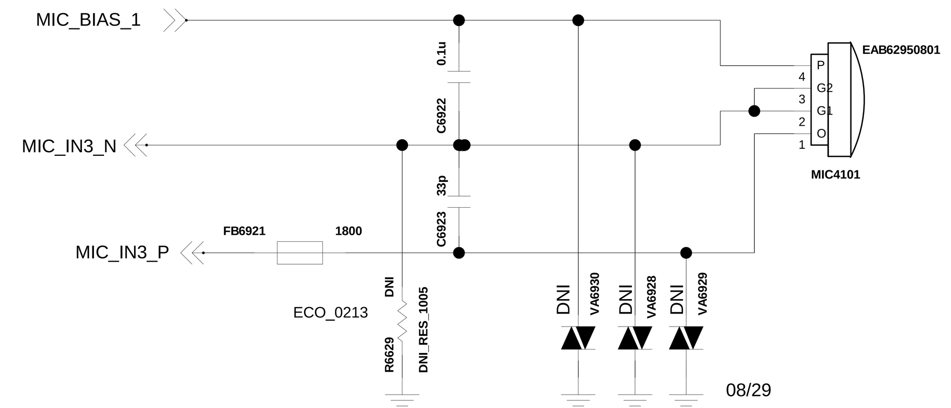
A2	PDM NUMBER	Rev
User Name	29 of 28	Drawing Date

< 6-9-2_2MIC >

Main MIC



Sub_MIC1

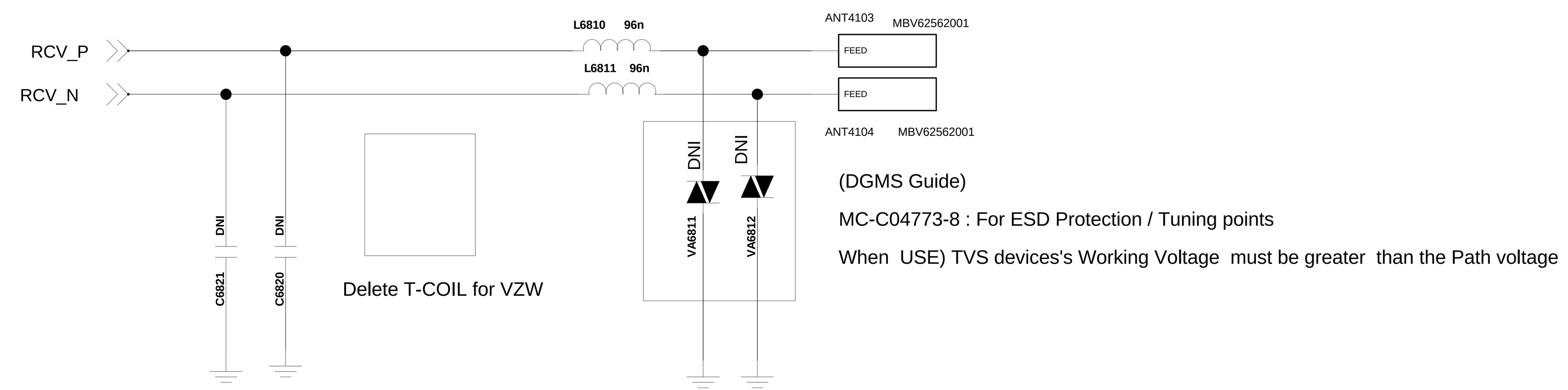


REV10_DNI varistor changed to 0 Ohm for camcording noise improvement. (03/28)

< 6-8-2_Receiver_TCOIL >

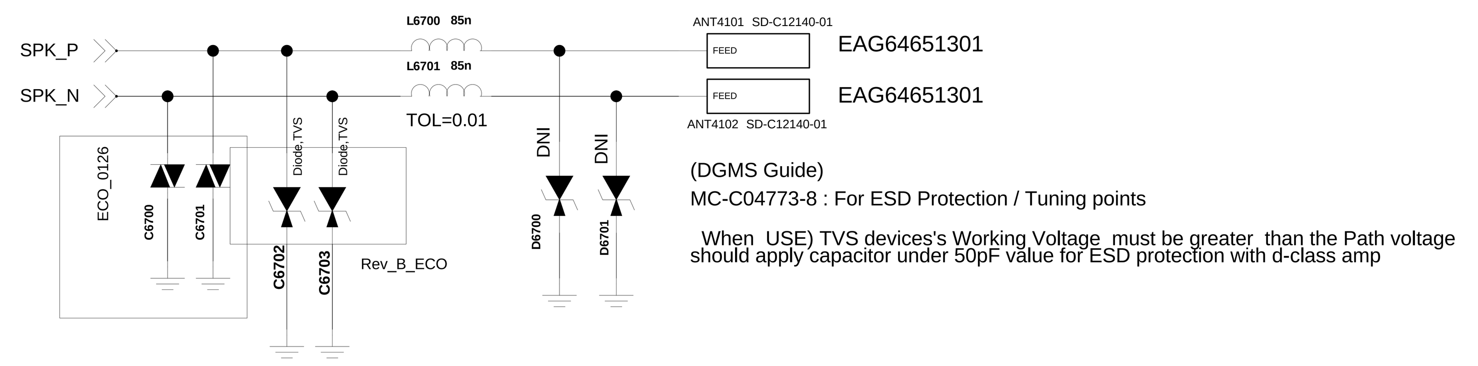
Rev_0.4

FPCB type : EAB64148501(no HAC, KIRIN)
FPCB type : EAB63688601(for TMUS)



(DGMS Guide)
MC-C04773-8 : For ESD Protection / Tuning points
When USE) TVS devices's Working Voltage must be greater than the Path voltage

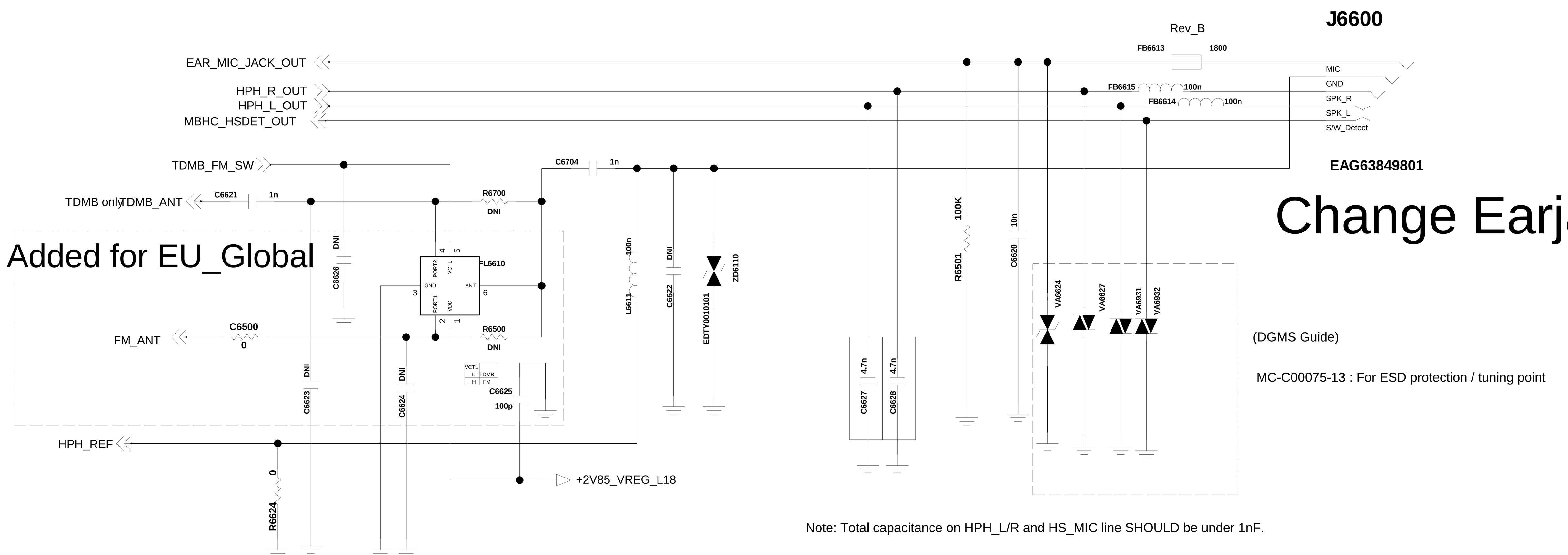
< 6-7-1_Speaker > Rev_0.4



(DGMS Guide)
MC-C04773-8 : For ESD Protection / Tuning points
When USE) TVS devices's Working Voltage must be greater than the Path voltage
should apply capacitor under 50pF value for ESD protection with d-class amp

< 6-6-1_Earjack > Rev_0.6

Circuit 2. Ear jack_3.5pi with MBHC



Change Earjack

(DGMS Guide)
MC-C00075-13 : For ESD protection / tuning point

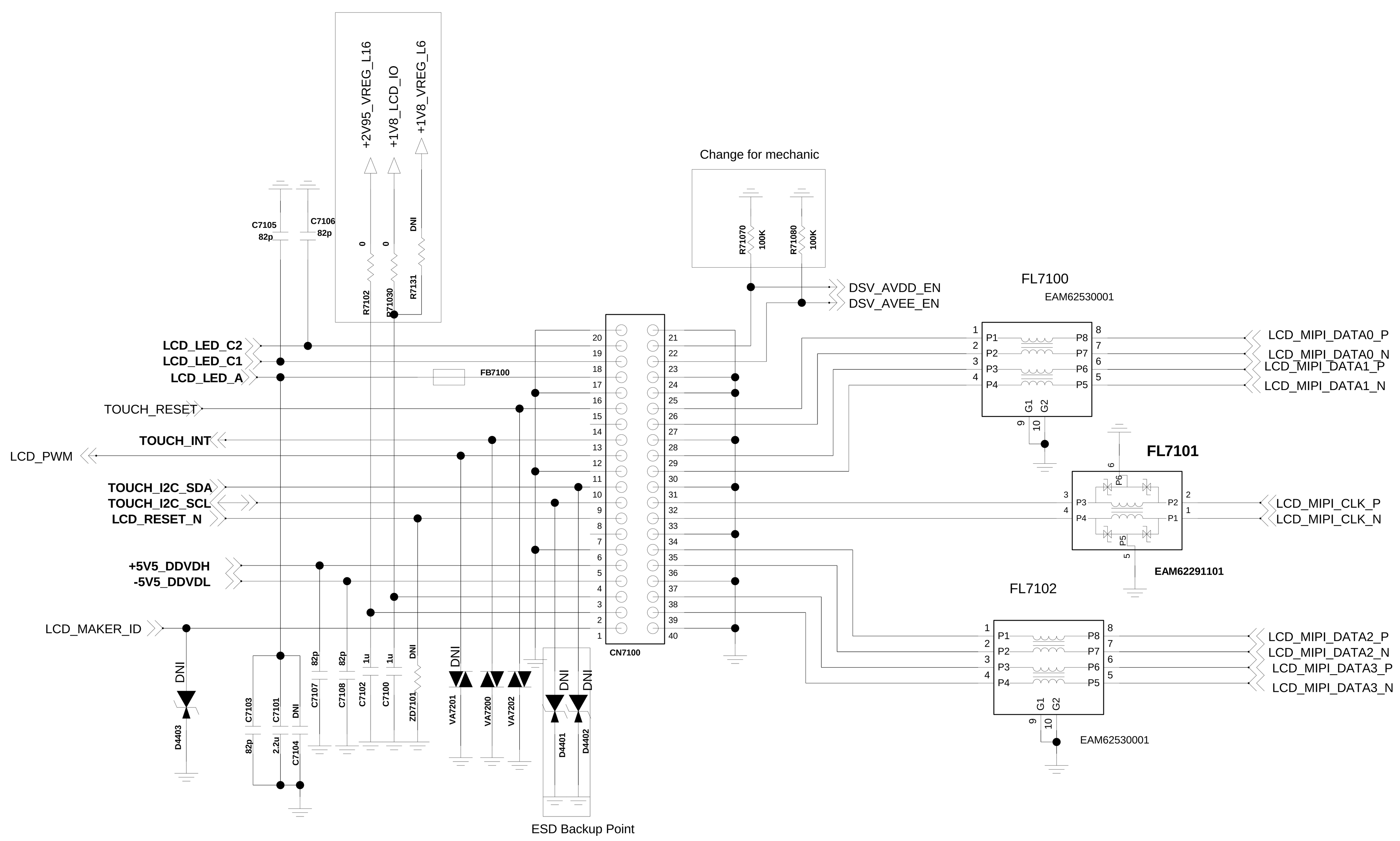
Note: Total capacitance on HPH_L/R and HS_MIC line SHOULD be under 1nF.

GND Close to EarJack

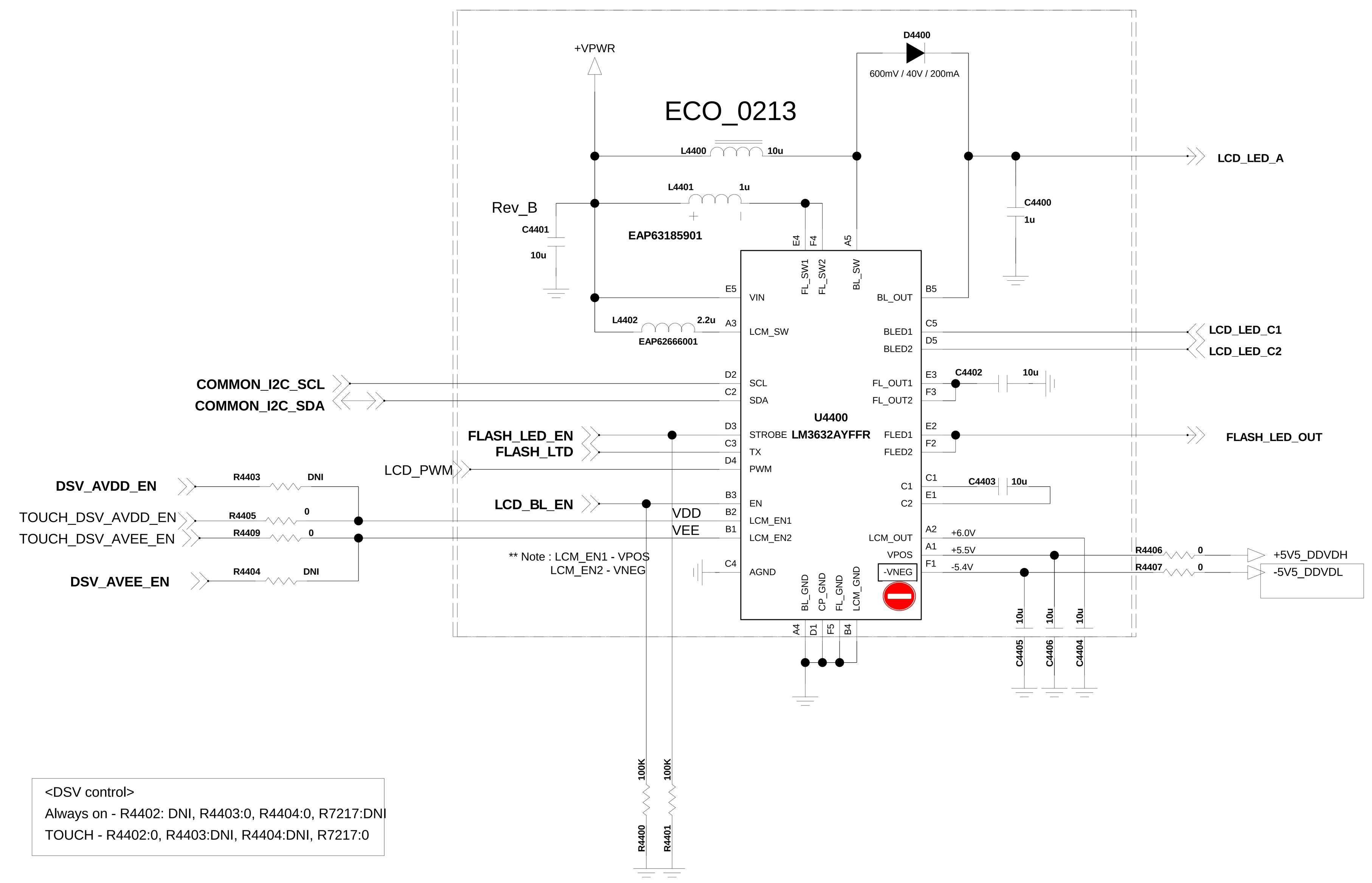
LGE Internal Use Only

LG Electronics
Schematic1
A2 PDM NUMBER Rev
34 of 31
User Name
Drawing Date

PH1_5.7" LCD Connector



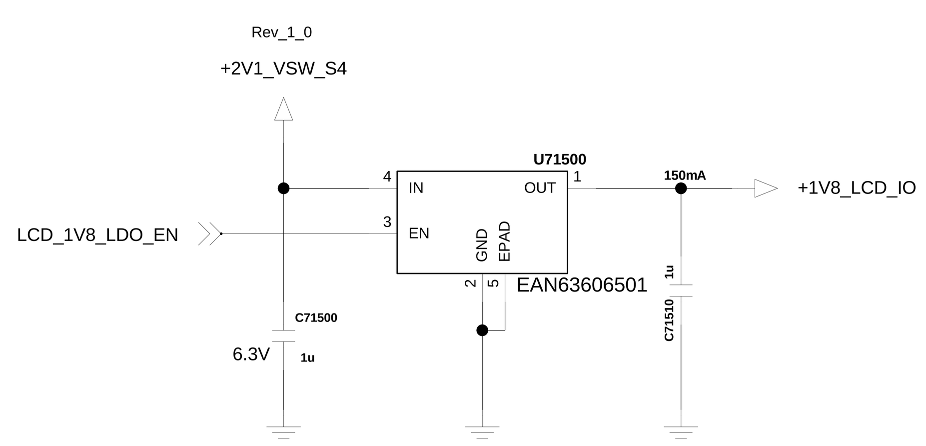
<4-4-7_LCD_Backlight_Flash_DSV_IC_LM3632> Rev_0.3



Delete LCD LDO spare

<Flash_LED>

LCD IO LDO



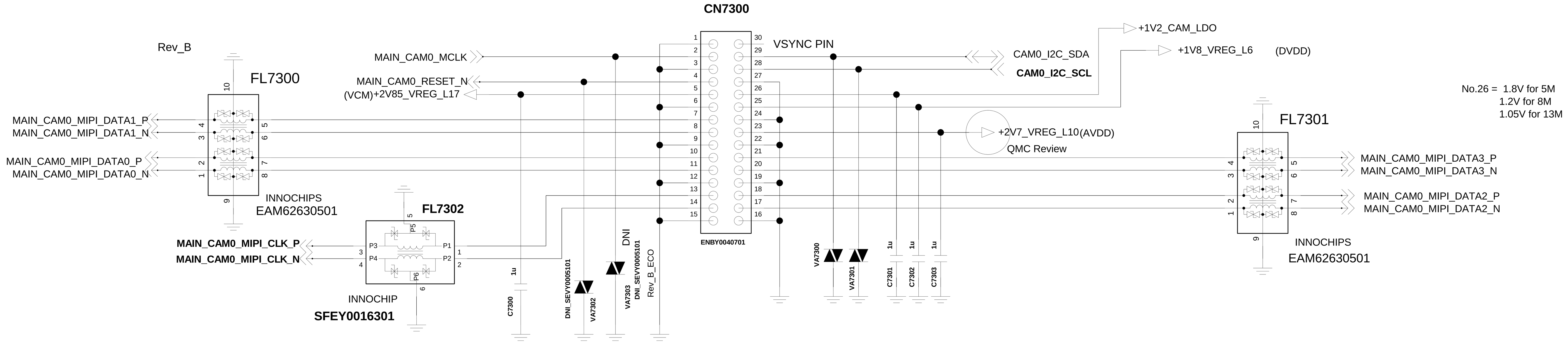
LGE Internal Use Only

LG Electronics	
TITLE	MODEL NAME
SIZE A2	PDM NUMBER
SHEET 32 of 29	Drawing Date
DRAWN BY	User Name

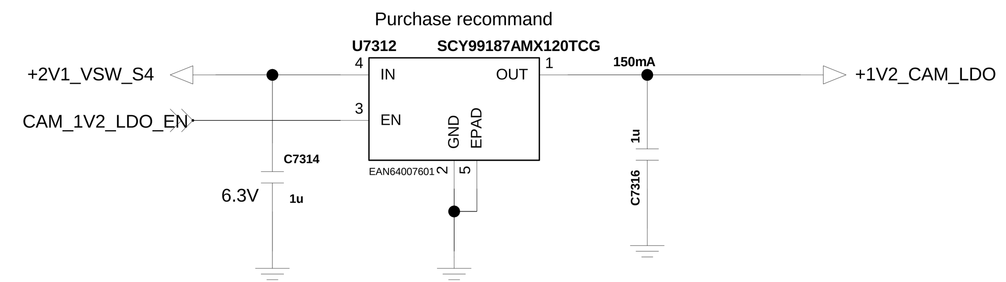
<7-3-5-1_30pin_5M_8M_13M_AF>

13 Mega camera module : EBP62702201 Rev_0.3

Check the pin map



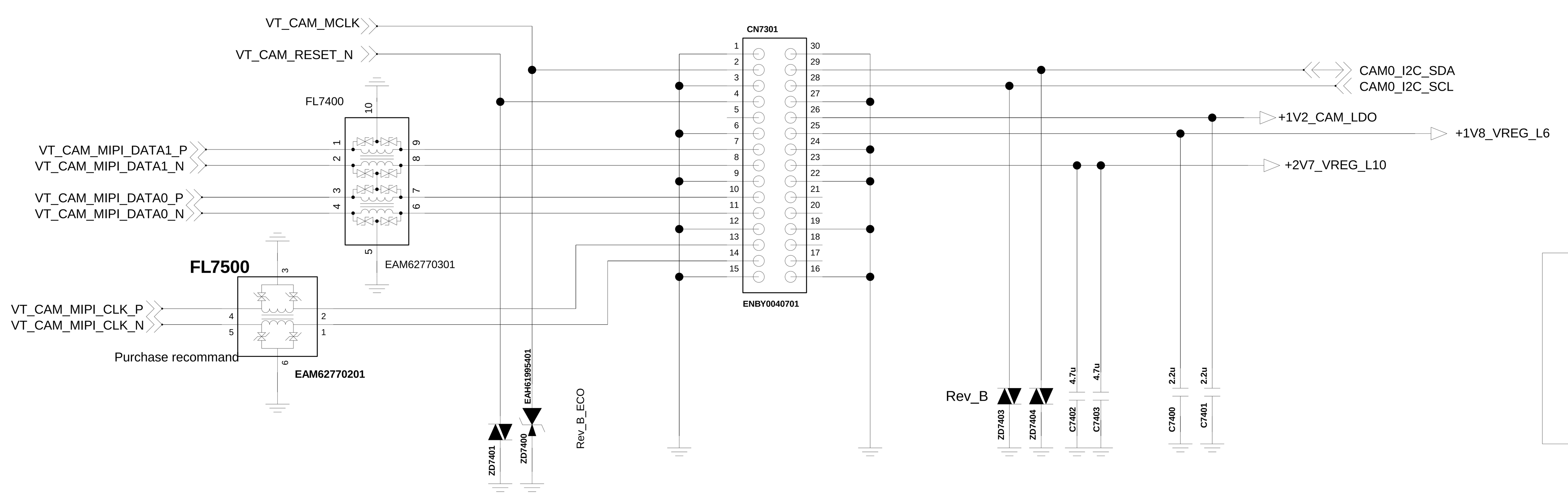
CAM LDO 13M&8M



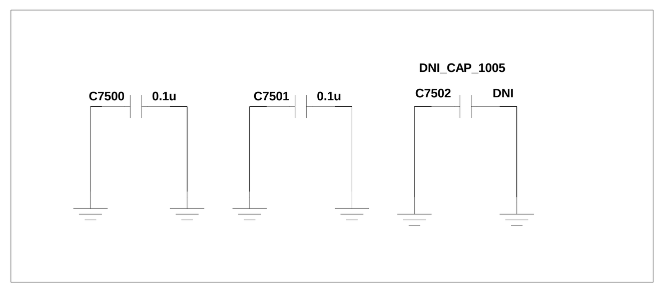
< 7-4-x_Front_Camera_30pin_5M_FF>

[DGMS Guide]

MC-C00081-17: 1. Add ESD Protection curcuit for Camera signal lines



Added Cap. for ESD



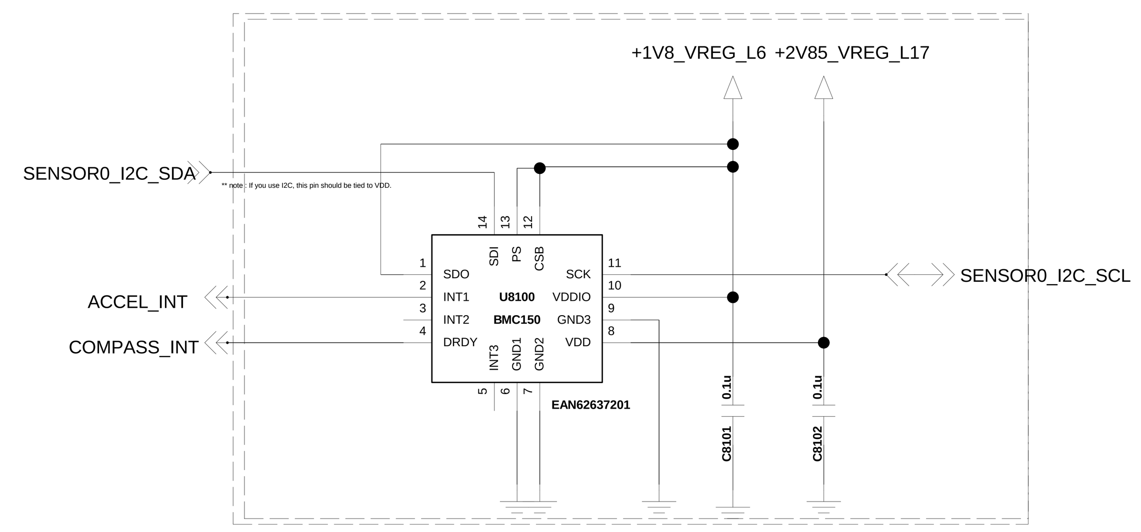
LG Electronics

MODEL NAME

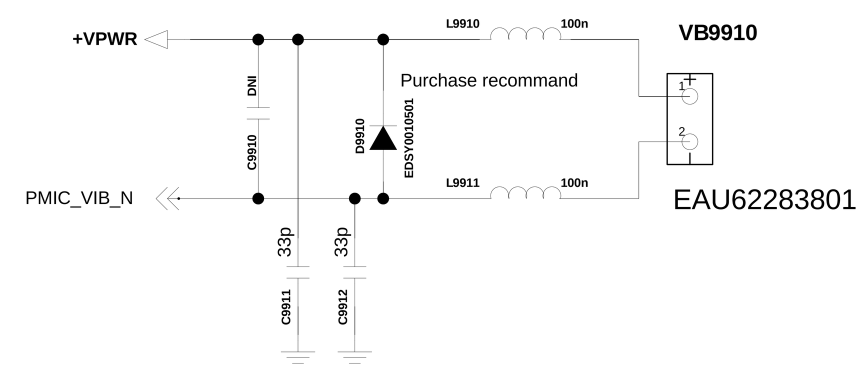
PDM NUMBER

A2 | Rev | User Name | Drawing Date

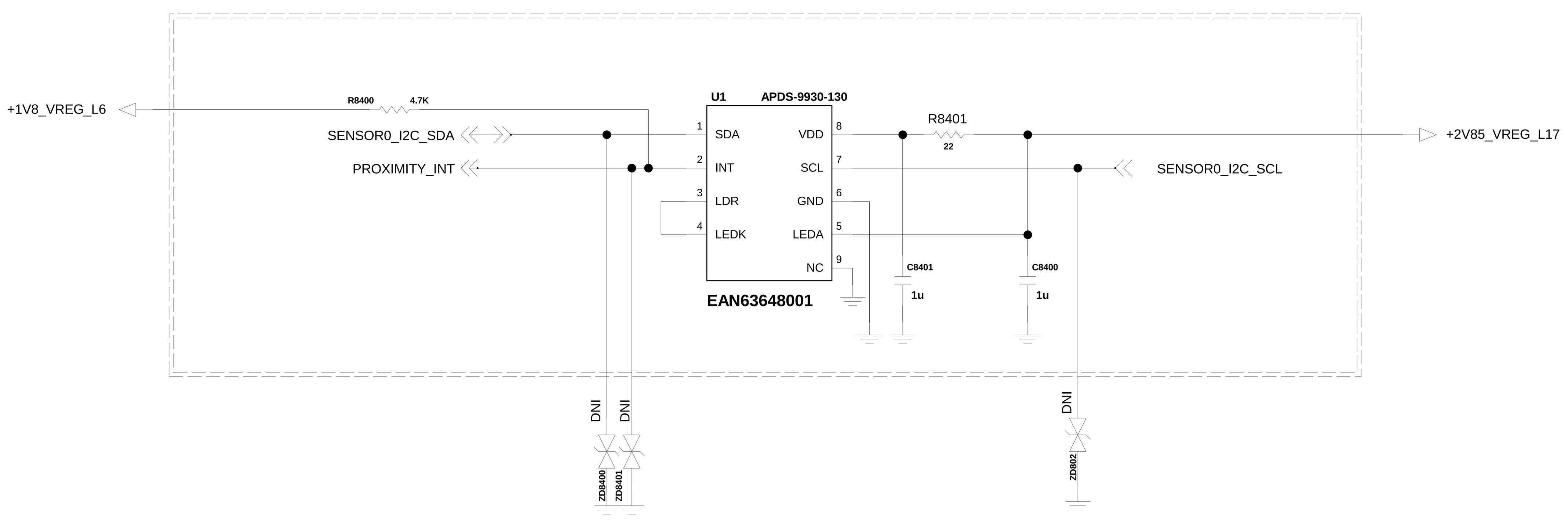
<8-1-2-2_Accel_Compass_BMC150> Rev_1.0



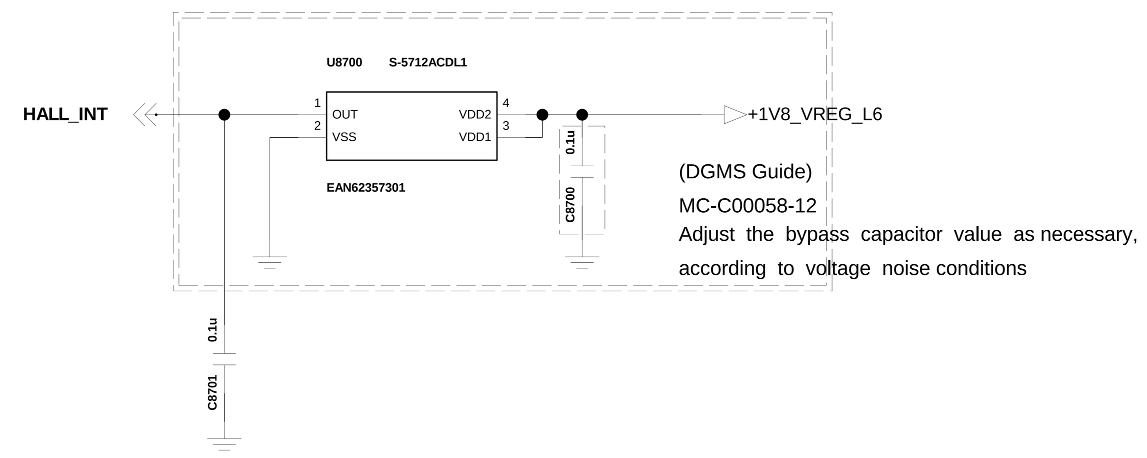
< 9-9-1-1_Motor > Rev_0.3



<8-4-1-1_Proximity_APDS-9930> REV.1.0



<8-7-1-6_Hall_IC_BU52033NVX> Rev_1.0



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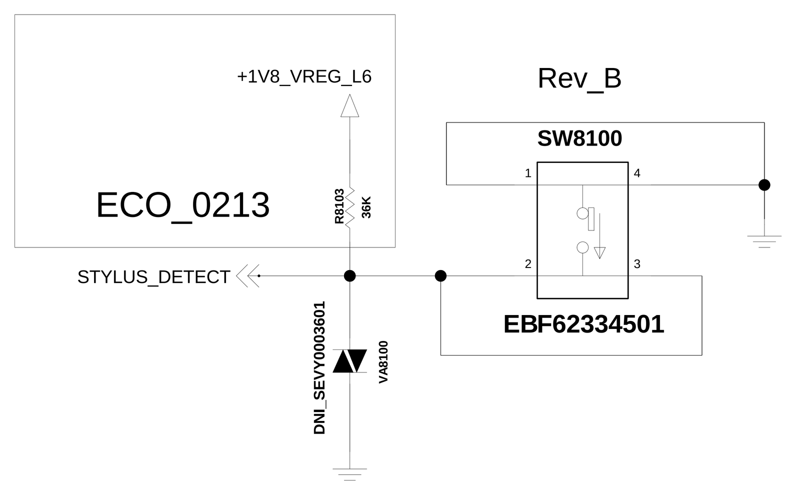


LGE Internal Use Only

LG Electronics

TITLE Schematic1

A2 PDM NUMBER Rev

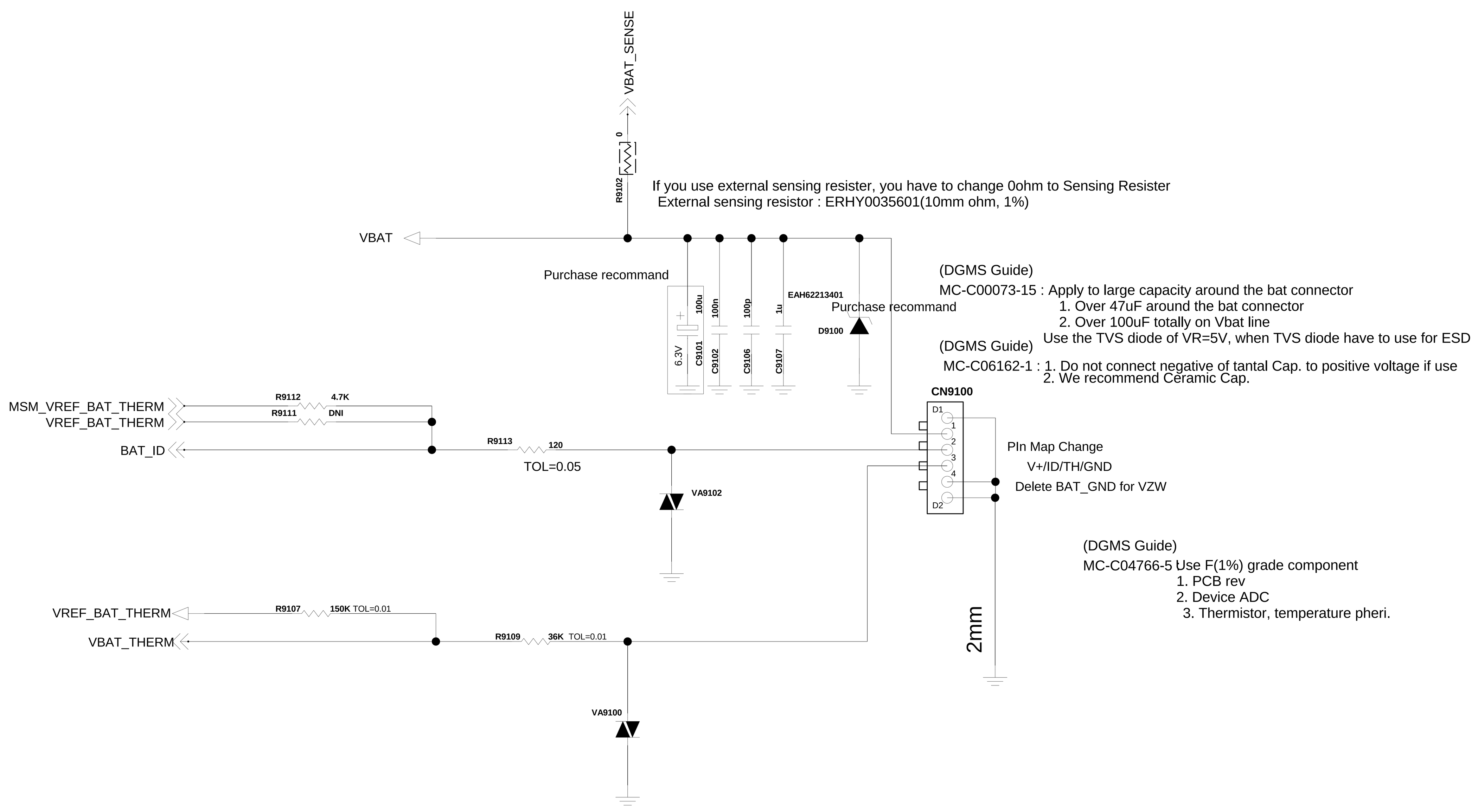
User Name Drawing Date

12 of 5

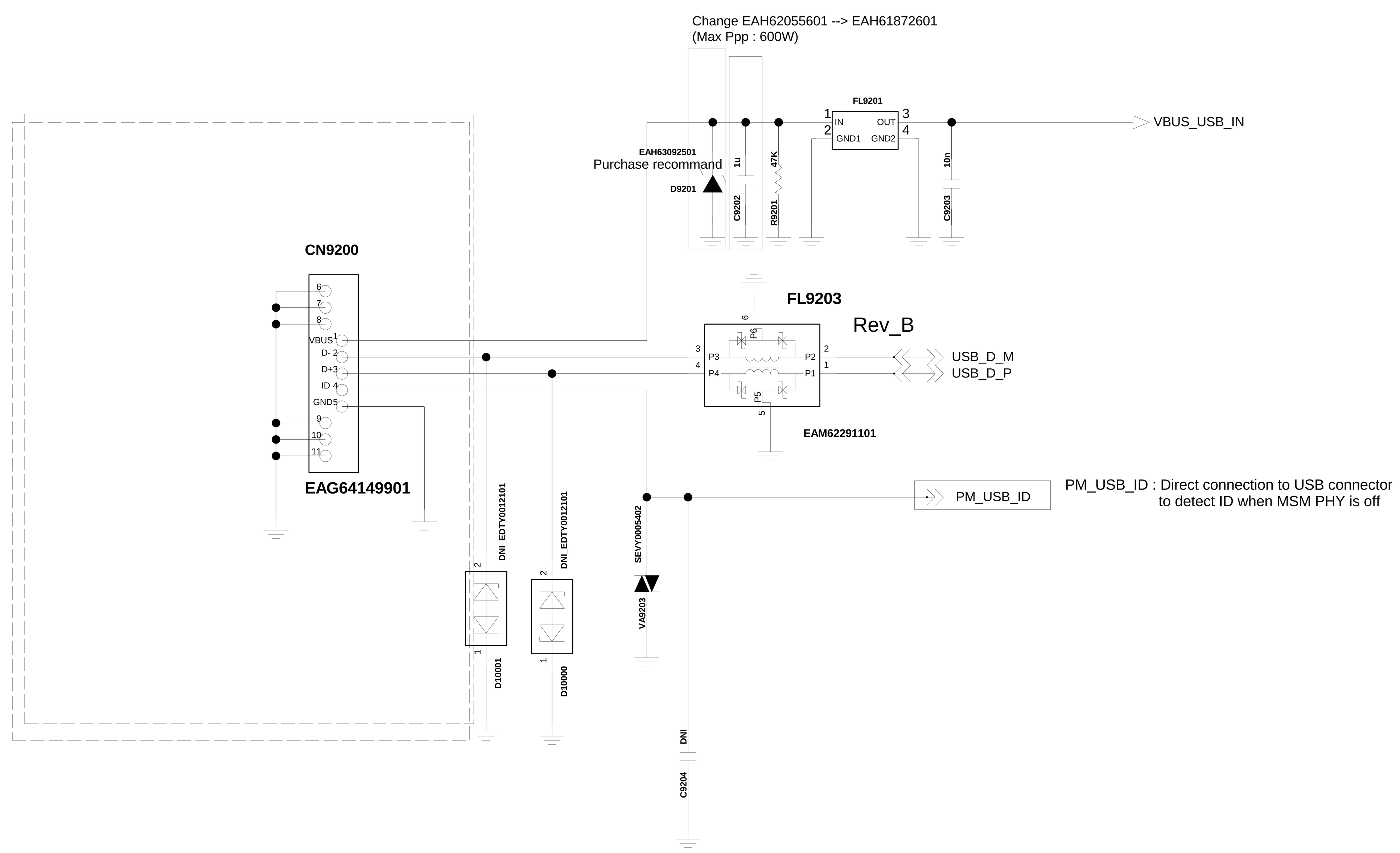
<9-1_Battery_Connector_4Pin>

<9-2-1_USB_IO> Rev.1.0

Circuit 1. Batt Conn. /wo EMI Filter



Connector 1. USB 2.0



LGE Internal Use Only

LG Electronics

TITLE MSM8926_X5_VZW

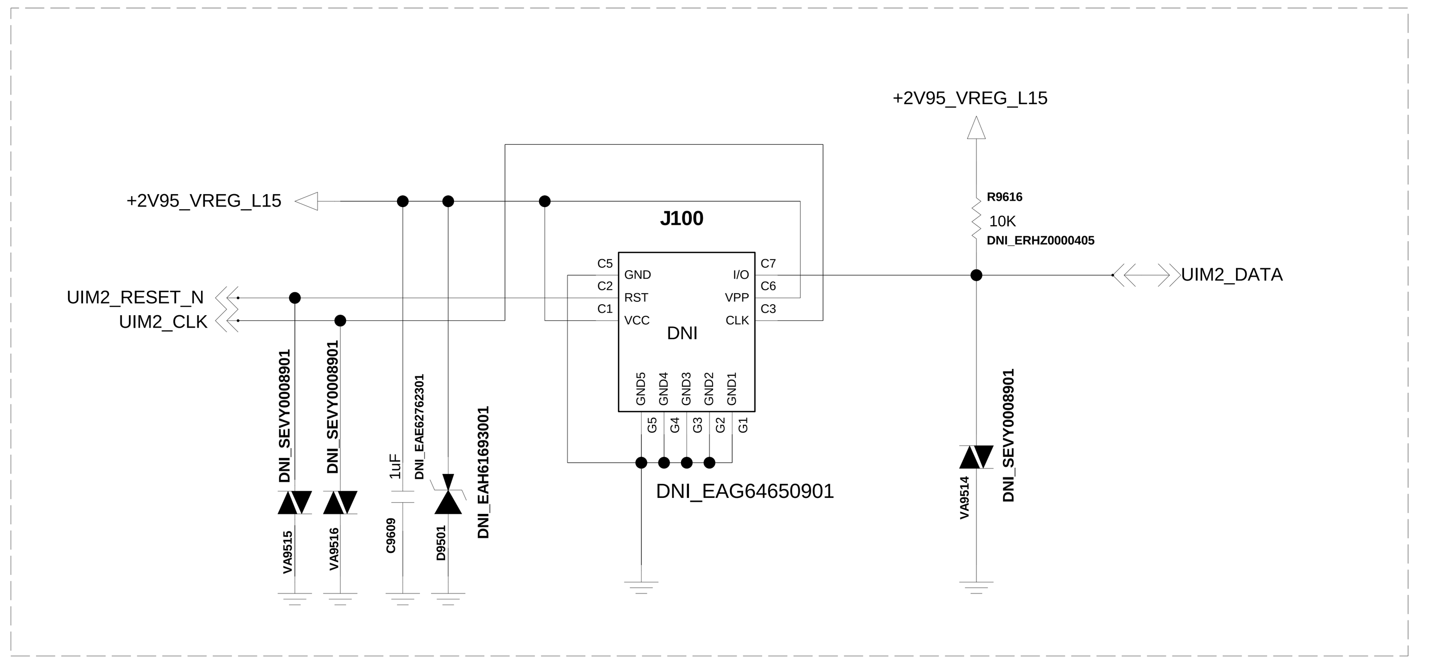
SIZE A2 PDM NUMBER Rev

SHEET 50 of 38

DRAWN BY User Name Drawing Date

Added for EU_Global

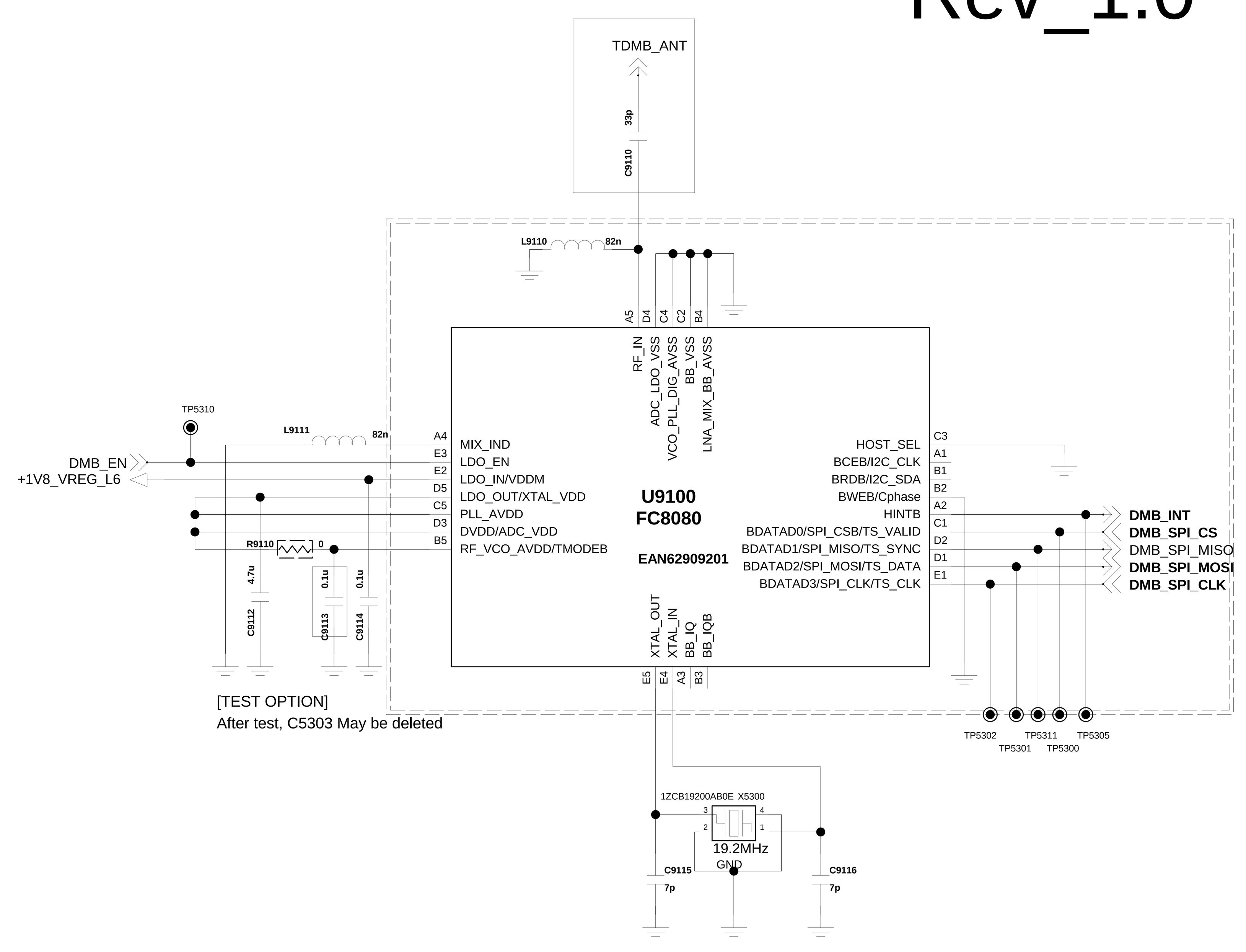
Dual Nano SIM(EAG64650901)



< 5-3-1-5_TDMB_FC8080 >

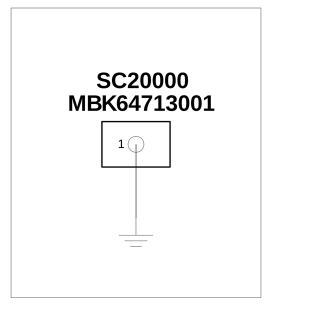
Rev_1.0

EAR_ANT



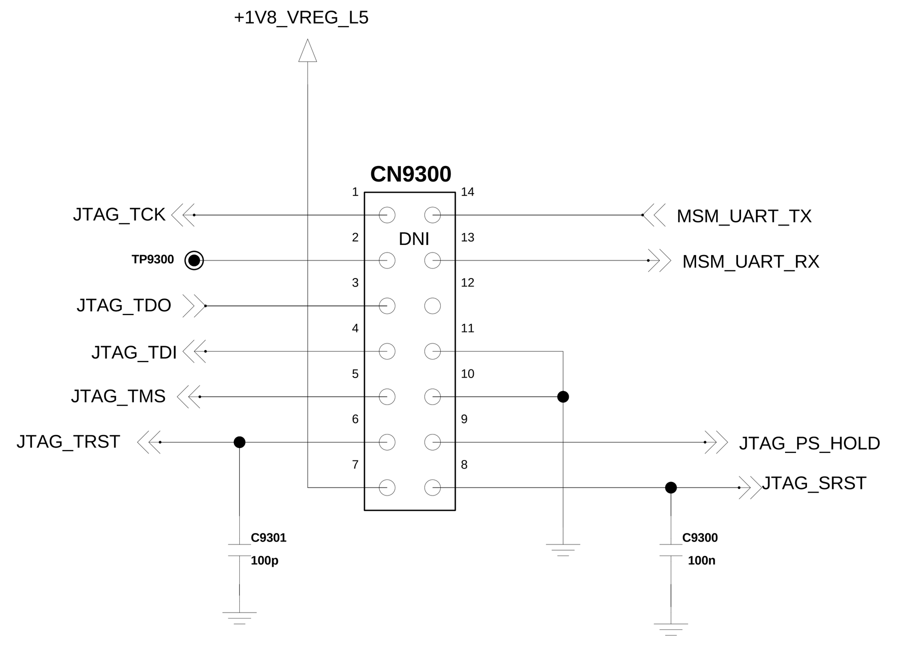
[TEST OPTION]
After test, C5303 May be deleted

TDMB ShieldCan



< 9-3-1_JTAG >

Rev_0.4



< 9-6_u_SDCARD_Socket > Rev.1.1

Circuit 1. u-SDCARD SIM Combo for QMC /w Low Detection

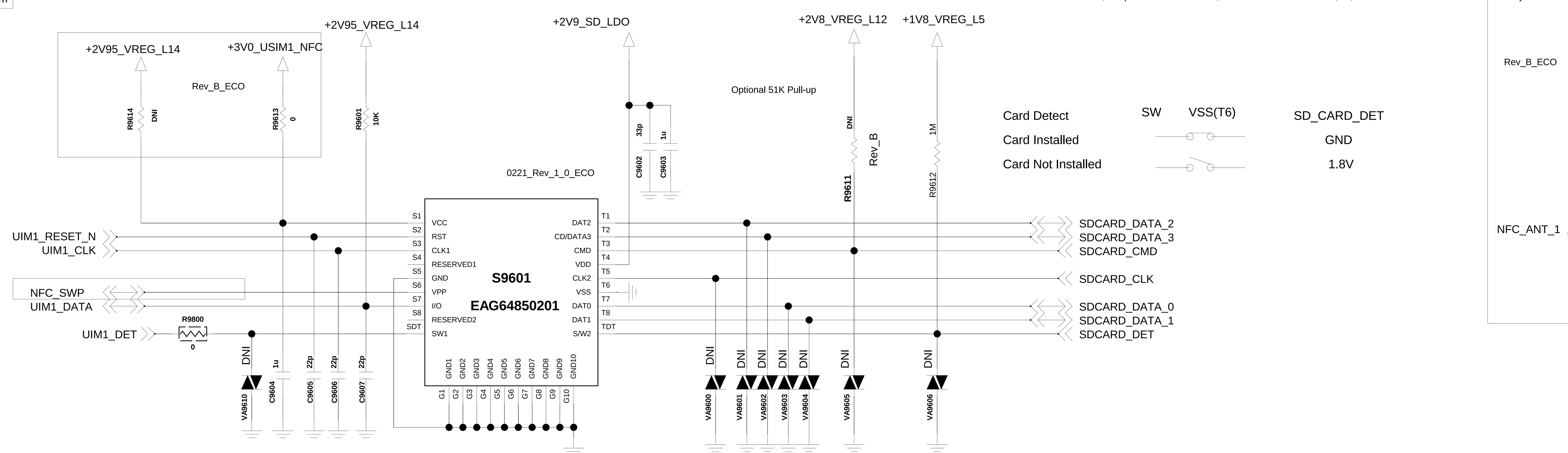
	R9614	R9613
NFC (X)	0 ohm	DNI
NFC (O)	DNI	0 Ohm

Added for NFC

Note 1. +2V95_SD_PULL_UP (1.8 or 2.95V) switches automatically per inserted SD Card version
For example, SD card Version 2.0 = 2.95V, Version 3.0 = 1.8V
Note 2. +2V95_SD_VCC is SD/MMC Power

Caution ! - According to MSM8960, APQ8064 BackPowering issue, +2V95_SD_PULL_UP was Always on during sleep mode.
However, It is possible for MSM8974, MSM8x10 .. to be turned off, so, FET circuit to turn off it is not needed any more.

Card Detect	SW	VSS(T6)	SD_CARD_DET
Card Installed			GND
Card Not Installed			1.8V



Nano SIM Combo Socket

(DGMS Guide)
MC-C04581-13 : Below items should be satisfied
1. SIM_DATA pull-up resistor
2. Use under 33pF (Cap_Varistor) on each signal line
3. When routing, SIM_DATA, SIM_CLK should be seperated
4. CAP,Varistor should Place close to connector

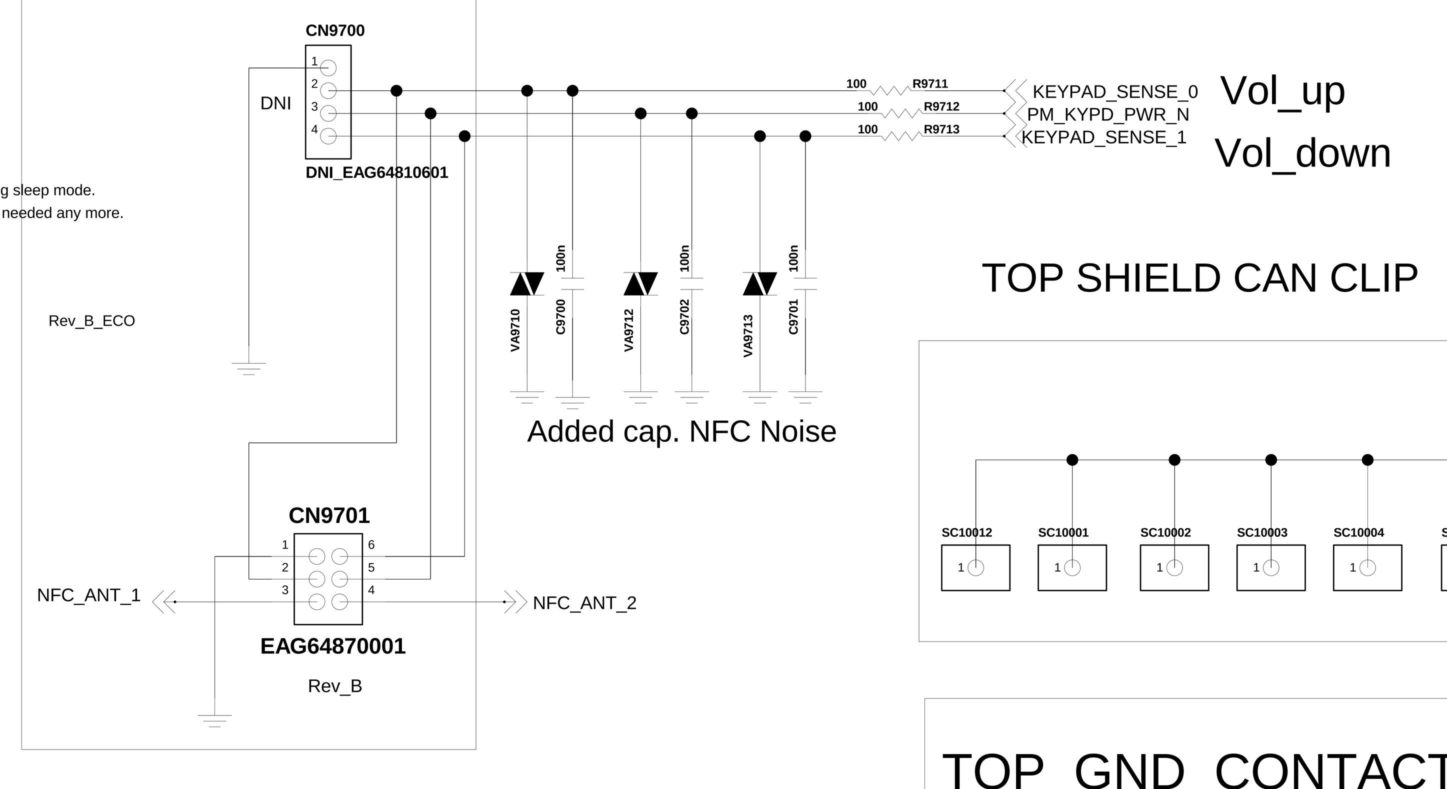
(DGMS Guide)
MC-C04582-14 : Below items should be satisfied
1. SD_DETECT Pull-up or Pull-down
2. Data/CMD line should be needed Pull-up resistor
(If chipset has internal Pull-up, comment it on circuit)
3. Use under total 10pF Capacitance on each line ; Data [0:3], CMD - Cap. TVS diode, varistor
(on other line, We recommend 10pF capacitance and use under 30pF capacitance)
4. CAP,Varistor should Place close to connector

(DGMS Guide)
MC-C05446-7 : Below items should be satisfied
1. Power of SD card should use independant Power source just like LDO/PMIC which can turn off it.
2. Per chip type which support SD card should manage current capacity.
(In case of High speed mode, SD Card Power should get margin of 200mA)
3. Do not connect serial component to SD card power, voltage drop can arise at Power line.

Circuit 3. u-SDCARD Straight /w High Detection

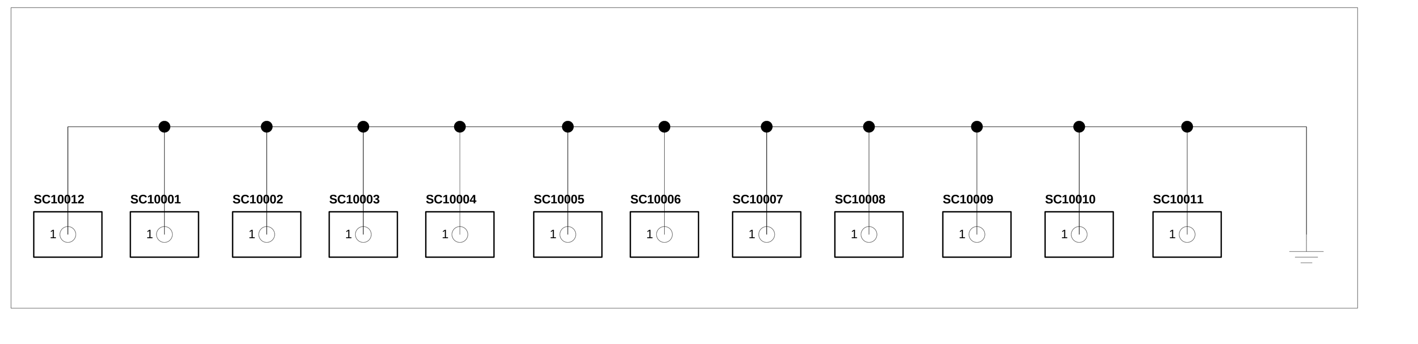
BACK_KEY & Flash LED

Added for NFC



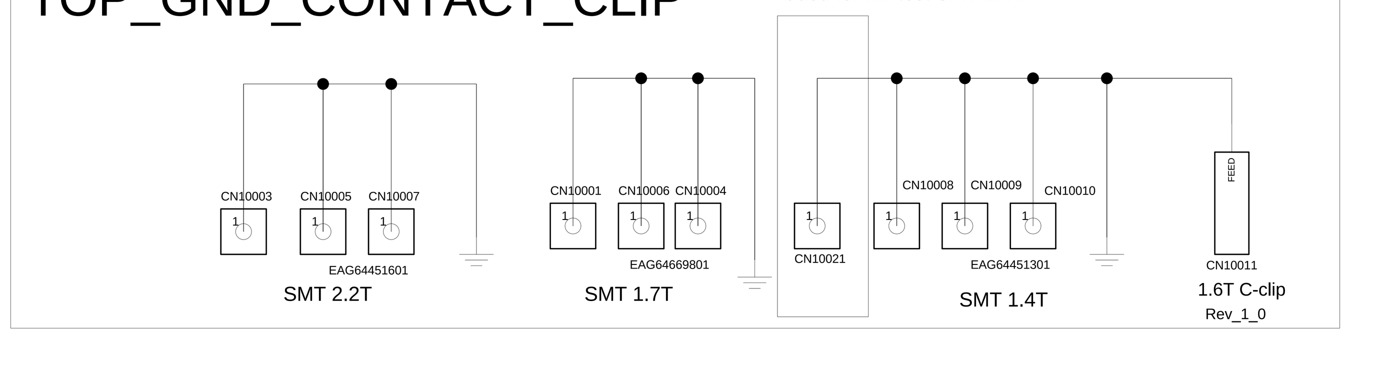
Added cap. NFC Noise

TOP SHIELD CAN CLIP



TOP_GND_CONTACT_CLIP

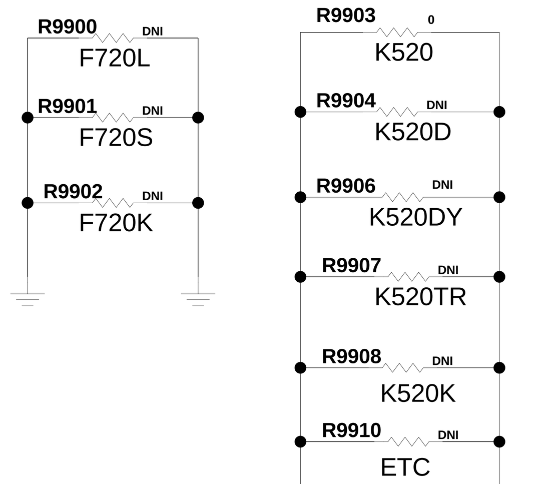
Added for RF test SMT 1.4T



SMT 2.2T SMT 1.7T SMT 1.4T

1.6T C-clip
Rev_1_0

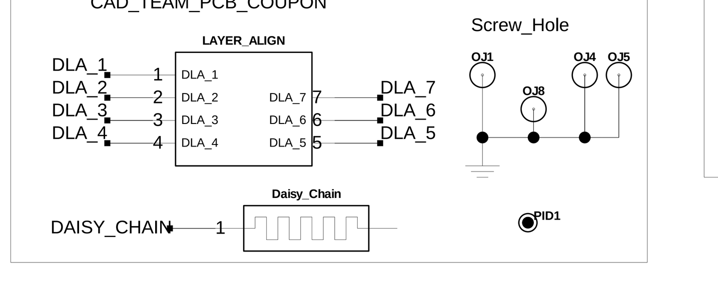
Domestic EU_Global



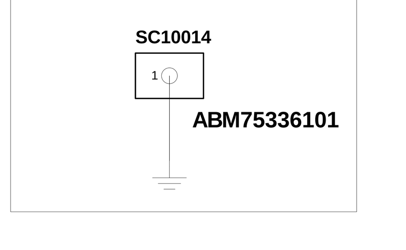
Model Index

	F720L	F720S	F720K	K520	K520D	K520DY	K520TR	K520K	ETC
R9900	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R9901	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI	DNI
R9902	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI	DNI
R9903	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI	DNI
R9904	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI	DNI
R9906	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI	DNI
R9907	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI	DNI
R9908	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm	DNI
R9910	DNI	DNI	DNI	DNI	DNI	DNI	DNI	DNI	0 Ohm

CAD_TEAM_PCB_COUPON



ECO_0126
SMT PMIC Shield CAN



LG Electronics
MODEL NAME
A2 PDM NUMBER Rev
User Name Drawing Date
1 of 2

LGE Internal Use Only

1,U1500_WTR4905_RF Tranciver(TOP)

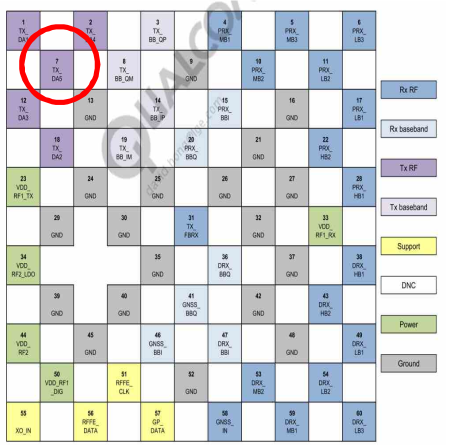


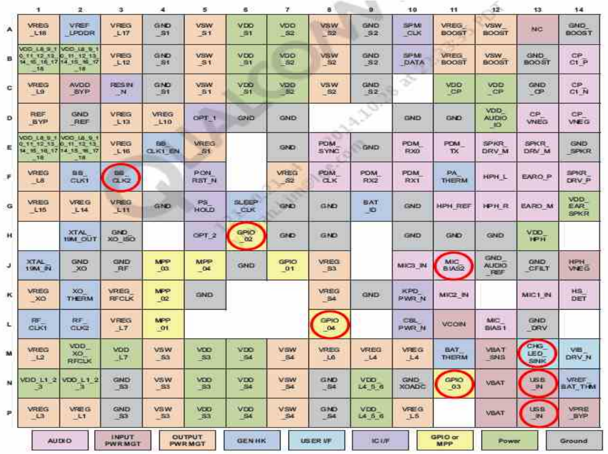
Figure 2-1 WTR4905 pin assignments – top view

O Not Used

1,U5100_WCN3620_BT/WiFi/FM IC (Top)

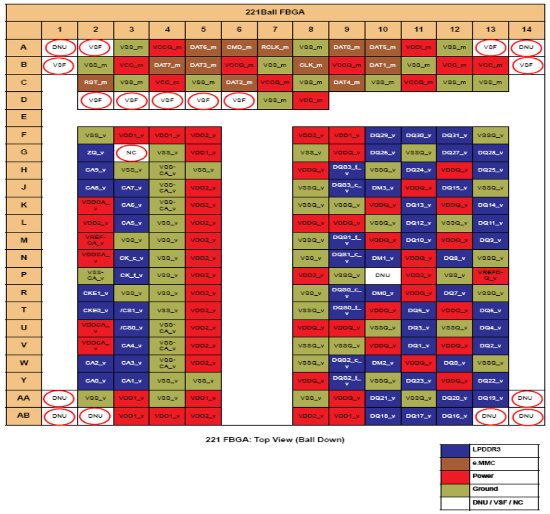
O Not Used

1. U4100_PM8916_IC,PMIC (BOT)



O Not Used

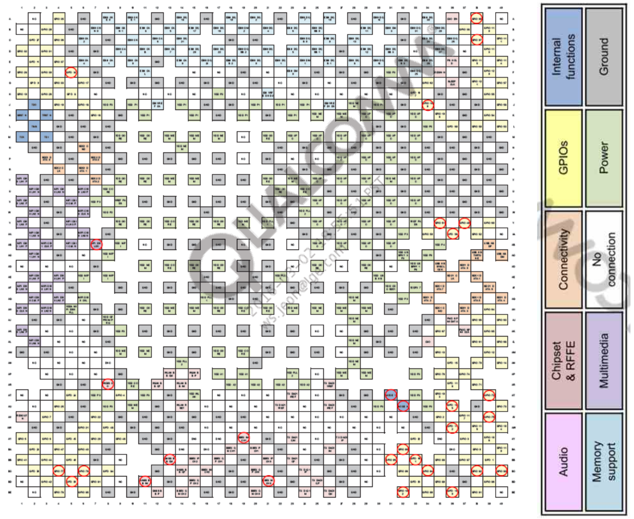
1. U3300_MCP+eMMC, 1.5GB/16GB (Top)



221 FBGA: Top View (Ball Down)

O Not Used

1. U2100_MSM8916_IC,Digital Baseband Processor (Top)



O Not Used

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